Brokers Who Do Not Bridge: When Assessment, Not Access, Shapes Broker Advantage

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Abstract

Structural theories describe economic brokerage as an opportune but fragile position bridging gaps between otherwise disconnected actors. In that view, brokering can undermine trust or eliminate the gaps that generate its value, and brokers need supporting institutions or deeper interpersonal ties to stabilize their role. Yet many corporate actors in positions of brokerage, from Visa to Amazon, have risen in market power, profit, and prominence even as networks transformed around them. When can brokering strengthen rather than undermine a broker's advantage?

I reconceptualize brokerage as outsourced relational work, in which a broker who assumes no inventory risk constructs viable matches between parties who cannot easily find or evaluate one another. Repeated matchmaking generates an informational byproduct that the broker can accumulate: knowledge of what makes pairings successful across the market, unavailable to clients who observe only their own dealings. An agent-based model of a matching market identifies conditions under which this informational advantage emerges and persists. Brokerage can provide assessment rather than access: clients outsource matching even when they can reach counterparties themselves, because the broker's predictions are valuable. Counterintuitively, the broker can become highly central in regimes where it engages in little bridging behavior: broker centrality is a product, not a primary source, of matchmaking quality.

While the model explores the dynamics of the broker's informational advantage, I theorize how this advantage can then enable the broker to become a principal selling the resource it once intermediated or monetizing data and analytics. This is *transient brokerage*, a process that highlights a broker's growing power rather than its fragility.

I Introduction

Intermediation in economic relations—the intervention of a third party to mediate or facilitate exchange between others—can take many forms, including that of brokerage. Brokers are intermediaries that do not assume inventory risk: they act as matchmakers, connecting others.

Brokerage is pervasive in a wide variety of markets and remarkably persistent across time and space. Brokers find themselves in an advantageous but delicate position. Unlike dealers and wholesalers, brokers do not sit on an inventory that they need to manage and can liquidate in case of difficulty. Unlike market makers, they keep offers and price information private (Rust and Hall 2003). They operate at the intersection of social groups (Burt 1992, 2004) and rely on secret information. As a result, they often lose the trust and legitimacy needed to continue their activities. Prominent theories of brokerage have described it as an opportune but inherently fragile role that often gets "stabilized" to avoid collapsing, through the social isolation of brokers, their organizational grafting, or their capture by one side (Stovel, Golub, and Milgrom 2011).

If that is the case, then how have so many corporate actors in positions of brokerage, from Visa to Amazon, managed to rise in market power, profit, and prominence even as networks transformed around them and could have undercut them?

Recent work by Ronald S. Burt, building on the tradition of structural-hole brokerage theory, has shown one important reason why the broker fragility narrative fails: many bridging ties are not weak and fragile, but rather strong high-trust ties forged through interpersonal history, and they provide a robust and enduring advantage to the broker who relies on them (Burt 2026; Burt and Opper 2026). However, this is not the full story.

Some brokers leverage their role to become principals rather than remain only intermediaries. In so doing, they stabilize their advantage and consolidate power while brokering becomes secondary or even disappears entirely. I call this *transient brokerage*, a phenomenon which current theories of brokerage do not satisfactorily predict or explain. Transient brokerage highlights how brokering sometimes declines or disappears because of the broker's power, not its fragility.

To explain transient brokerage, this article redefines brokerage as *outsourced relational work*: the broker constructs viable matches between parties who cannot easily find or evaluate each other. This relational work generates an informational byproduct that the broker can leverage. Structural position provides the access that feeds learning, but while each successful match between previously unconnected parties can erode the broker's structural advantage by densifying the network (Burt 2000, p. 356), it also strengthens the broker's informational position. The broker accumulates knowledge of what makes pairings successful.

In that way, the broker converts structural capital into informational capital through the act of brokering. This is particularly true for corporate brokers, who operate at scale and whose relational work benefits from legal privileges conferred by the corporate form. Depending on the market in which the broker operates, the nature of the matching problem, and the relations between the actors involved, the broker's information advantage can support a transition from intermediation to capture. The broker dismantles the relational package that was negotiated with its principals and converts vicarious relational work into personal economic advantage. If successful, **the broker becomes a principal selling the resource it once intermediated or monetizing data and analytics**. *Corporate* relational work is a particularly advantageous basis for such capture. This explains the rise in prominence of some firms acting as brokers

This article shows how the broker builds and sustains an informational advantage, through social theory illustrated by computational simulations, and explains how this advantage can then enable the broker to transition from intermediary to principal. It attempts to answer three questions about brokerage, generating propositions to be tested in future empirical work:

(i) What value does a broker provide?

(ii) How does brokering alter its informational and structural position?

(iii) Under what conditions can the broker convert its advantage into resource capture or data enclosure?

I develop an agent-based model (ABM) of brokerage in a general matching market and use it to formalize and illustrate how brokers create and preserve an informational advantage through repeated matchmaking. The simulations show how several simple elements interact to produce non-obvious dynamics, and they complicate the expectation that brokers densify networks around them and erode their own position. The results suggest that brokerage can provide assessment rather than access and that clients continue outsourcing even when they can reach counterparties themselves. A broker's structural prominence can be produced endogenously by its success in the work it performs, centrality is separate from actual bridging behavior, and a broker need not liquidate its own structural position. A broker can be highly central while engaging in little to no bridging. The results recast betweenness centrality not as a neutral measure of a broker's strength grounded in network topology, but as a product of the kind and quality of its relational work.

The rest of this article proceeds as follows. Section II reviews structural and processual theories of brokerage and their blind spots. Section III reconceptualizes brokerage as outsourced relational work and traces how this work is transformed when performed at scale by corporate actors. Section IV presents the ABM and its results illustrating how a broker produces and sustains their advantage through this work. Section V builds on this foundation to theorize one possible trajectory that can follow from the broker advantage, transient brokerage, and the conditions under which a broker can become a principal. Section VI discusses scope conditions and implications for the political economy of corporate intermediation, and Section VII concludes.

II Current Theories of Brokerage Do Not Explain Capture by the Broker

Brokerage tends to emerge in markets where information is scarce or complex, participants are few, transactions are intricate, or institutions are poorly developed (Stovel, Golub, and Milgrom 2011). These conditions make participants hard to find, resources hard to value, or transactions hard to structure. They also make the intermediation work of a broker valuable. Real estate, mortgage, stock, and commodity markets are common examples of markets with many active brokers. Brokerage may also be helpful in circumstances where tight-knit, closed groups monopolize resources or information, and outsiders cannot easily access them (Stovel and Shaw 2012, p. 140). Brokerage can maintain or reinforce group boundaries, or it can blur or collapse them (Stovel and Shaw 2012), with consequences for social segregation, social integration, and relations between groups or communities (Galster and Godfrey 2005).

Past research has proposed two conflicting sets of definitions of brokerage (see Erikson 2013, for a review): narrower formalist or structuralist definitions (Burt 1992; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982), and broader processual definitions (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012).

A Formalist and structuralist definitions

Early brokerage research examined the structure of networks and the positions of actors to identify brokerage forms, categorize them, and determine their consequences (Burt 1992, 2004; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982; Simmel 1964, 1971). Brokerage opportunities arise when there are gaps in a social network, such that some nodes are close (in degrees of separation) yet disconnected (Burt 1992, 2004; Marsden 1982). Brokerage is synonymous with open triads or structural holes: a structure in which a third actor bridges two otherwise unconnected nodes. The third actor mediates flows of resources and information and facilitates transactions between others who lack access to one another (Fernandez and Gould 1994, p. 1457; Marsden 1982, p. 202). Researchers in this tradition generally set aside interactional content to achieve greater generality (Erikson 2013). For instance, Burt (1992, p. 36) was explicit in treating "motivation and opportunity as one and the same."

Burt (1992) operationalized the structural-hole concept, proposing the measures of network constraint and effective size to quantify how well positioned an actor is for brokerage, which allowed the theory to be tested across a wide range of settings. For instance, in a follow-up study, Burt (2004) investigated the network position of 673 managers in a large electronics company. He found that those who spanned structural holes generated ideas that senior management judged more valuable and were rewarded with better performance evaluations, promotions, and compensation. He suggested that brokers appear more creative because they can take insights that are commonplace in one group and introduce them to other people to whom they appear novel and original (Burt 2004, p. 388).

As such, brokers derive an advantage from their ability to exploit pre-existing information in the network. Because of their position, brokers get privileged access to diverse ideas from across the network. They can see opportunities that others do not, and they can strategically use their position by providing or withholding information (Burt 1992) or using it to sustain their own creative endeavors. Exposed to varied perspectives, they are well-positioned to build a richer understanding, synthesize across domains or outlooks, and develop a unique vision (Burt 2004).

Defining brokerage based on structural holes, however, excludes situations that clearly appear as brokerage to a lay observer. For instance, Alice wants to sell her house and asks a real estate broker to list it. Soon after the real estate broker puts up the 'for sale' sign, her neighbor Bob, whom she knows personally, contacts the broker to purchase the house. A broker intermediates the transaction, yet no structural hole existed between buyer and seller.

B Processual definitions

More recently, some researchers have recast brokering as a set of relational processes (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012). Stovel and Shaw suggested that brokerage was "the process of connecting actors in systems of social, economic, or political relations in order to facilitate access to valued resources" (Stovel and Shaw 2012, p. 141) (see also Stovel, Golub, and Milgrom 2011). Brokers exhibit two crucial features: they are positioned as bridges across gaps, and they help transfer resources or knowledge across those gaps (Stovel and Shaw 2012, p. 141). This definition remains constrained by social structure but requires brokerage behavior. Hence, there can be structural holes without brokerage—some bridging actors do not transmit anything across holes—but no brokerage without structural holes. In this way, it runs into the same conceptual problem as purely structural definitions.

Obstfeld, Borgatti, and Davis (2014), building on Marsden (1982), proposed a purely processual definition: brokerage is a process by which an actor "alters interaction between two or more parties in a wide variety of triadic structures" (Obstfeld, Borgatti, and Davis 2014, p. 136), meaning that he "influences, manages, or facilitates" the interaction (Obstfeld, Borgatti, and Davis 2014, p. 141). This eliminates any structural requirement beyond the need for three or more actors. It is no longer necessary for the alters to lack access to or trust in each other. There can be structural holes without brokerage, and there can also be brokerage without structural holes: brokerage behavior can, and often does, occur in dense networks.

Previous work on brokerage, particularly in the processual tradition, has also identified categories of brokers depending on the nature of their activities. Middleman, transfer, or conduit brokers facilitate economic transactions without necessarily altering the structure of a local network, like stockbrokers (Stovel and Shaw 2012, p. 146) (Spiro, Acton, and Butts 2013) (Obstfeld, Borgatti, and Davis 2014, p. 141). Catalyst, moderator, or matchmaking brokers focus on creating lasting ties, like headhunters and party hosts (Stovel and Shaw 2012, p. 146) (Obstfeld, Borgatti, and Davis 2014) (Spiro, Acton, and Butts 2013). Finally, *coordinators*, such as middle managers or disaster response teams, work to transform existing ties by reconfiguring local networks of interdependence (Spiro, Acton, and Butts 2013, pp. 131–132).

While definitions based on structural holes were very restrictive, Obstfeld and co-authors' processual definition is sweeping. For instance, they suggested that, under their definition, a manufacturing firm acts as a broker when it buys raw materials from an alter, transforms them into a product, and sells the finished product to another alter (Obstfeld, Borgatti, and Davis 2014, p. 145). Most triadic or polyadic relations, in this account, imply brokerage. Indeed, many descriptions of brokers in the literature are so broad as to invite the charge that a concept that explains everything explains nothing. The most varied social facts have been described as acts of brokerage: mediating conflict (Obstfeld, Borgatti, and Davis 2014), reselling stolen property (Steffensmeier 1986), advocating on behalf of someone, making sense of the world for someone, and helping connect epistemic communities (Stovel and Shaw 2012, p. 140), to name only a few.

C Shortcomings of existing frameworks

Work that relies chiefly on the structural holes idea generally makes clear that transferring knowledge and ideas is a key element of brokering and of the broker advantage. Yet, the information being considered is generally pre-existing, ambient information, not the one that the process of brokering generates (Burt 2004). Moreover, the structural account leaves the actual work of brokering largely implicit. Structural position indicates places where brokerage is possible, but not whether anyone is doing the work of brokering—work for which structural holes are neither necessary nor sufficient.

Meanwhile, broad processual definitions blur the lines between vicarious and personal relational work. They include under the same umbrella cases where the intermediary deals with others on its own account, and cases where the intermediary works on behalf of others. As such, this perspective doesn't examine the informational byproduct that results from repeated vicarious work. Information is often simply listed among the resources that can be transferred through a broker, along with material or financial resources (see e.g. Obstfeld, Borgatti, and Davis 2014, p. 141; Stovel and Shaw 2012, p. 141).

Neither perspective offers a cohesive theory of the role of information in brokerage, even though information shapes every stage of the brokering process. Before brokerage, the distribution of information shapes the opportunity or need for it: brokerage arises partly out of informational inequality, and the broker exploits what it knows that one or both sides do not. During brokerage, the transmission of information is part of the brokerage act itself, as the broker shares enough to make a match viable while withholding enough to preserve its advantage. After brokerage, the distribution of information has been altered, and the work has generated a byproduct the broker retains, which may shape the broker's opportunities. The circulation of information is at once a determinant of brokerage, a component of the work of brokering, and a consequence of brokerage.

This article reconceptualizes brokering as outsourced relational work and builds on this theoretical move to explain how brokerage sometimes weakens or disappears because a powerful broker captures resources or encloses data to become a principal.

III Outsourced Relational Work and the Broker Advantage

Social actors engage in *relational work* when they try to interpret and match social ties, economic transactions, and media of exchange, to create viable matches that will shape the outcome of their interaction (Zelizer 2012, p. 151, 2005) (see also Bandelj 2020, p. 255). These matches and their shared meanings form distinct *relational packages*. For instance, in a marketplace, actors trading directly with one another draw on cultural knowledge to reach a shared understanding of their situation and the 'rules of the game,' and to agree on terms, including price (Block 2012, p. 137).

For the broker's clients, brokerage is *outsourced relational work*. For the broker, brokerage is *vicarious relational work*. In economic markets, this outsourced relational work is usually performed at scale by a *corporation*. This is more than a question of definitions: it shapes the broker's opportunities, its strategic advantage vis-à-vis its clients, and its trajectory.

A From personal to *outsourced and vicarious* relational work

In brokered transactions, brokers take on the work of constructing a shared reality for limited purposes. The core of what brokers do is relational matching (see Zelizer 2012). Brokers are characterized by the relational work they do *on behalf of others*.

For the broker, this vicarious work requires another layer of relational work to negotiate brokerage relations with its clients. Brokerage corresponds with a distinct relational package. Not all intermediation is brokerage. In the relational package of economic brokerage, the broker performs a matching service for others, usually for a fee, commission, or retainer. Brokers bundle various services and amenities as part of that package: they take on the efforts of search and negotiation, they provide access to otherwise inaccessible counterparties, they share with their clients their expertise and the benefits of the private information they hold about quality, pricing, and availability, and they document and facilitate transactions, which often require logistical, bureaucratic, or regulatory work.

The middleman, catalyst, and coordination brokerage described in Section II are not exclusive categories of brokers, but rather, components of the work of brokering, which involves a bit of each in varying proportions. An important part of brokers' relational matching is to facilitate interaction (middleman brokerage), but this work also directly impacts interpersonal relations and local networks, often in an enduring way, by creating new and lasting ties (catalyst brokerage) or by reorganizing local networks (coordination brokerage).

Brokerage inherits the dual character of relational work, which can both foster and reduce informational asymmetry when constructing shared understandings while maintaining strategic opacities (Block 2012, p. 138). In performing relational work on behalf of others, the broker simultaneously clarifies and obscures: it transmits enough information to make a transaction legible and a match viable and conceals enough to justify its fee and preserve its advantage. The same opacity can do social as well as economic work. Rossman (2014) shows how brokerage functions as obfuscatory relational work that manages the taboos around disreputable exchange. Outsourcing relational work provides cover to parties in morally fraught matches. What the broker withholds is also the raw material of its later advantage: information it keeps from circulating, accumulates, and, under the right conditions, eventually monetizes.

Vicarious work relates to entities or resources that are *not* under the broker's personal ownership and control, and the shared understanding of the parties is that the broker is *not a principal* in the brokered transaction (see e.g. Stovel, Golub, and Milgrom 2011, p. 21326). Brokers generally have limited personal involvement in brokered interactions beyond their role as facilitators. The broker is a supporting actor, not a main character. Intermediaries who have full control over traded resources occupy a very different position in terms of risk, power, and vulnerability.

Another way to put it is that in economic markets, brokers are intermediaries that *do not assume inventory risk.* That is, if Bob helps Alice and Charlie conclude a sale, Bob is a broker. If Bob buys from Alice and eventually leases to Charlie, he is not. For instance, a real estate broker using her market knowledge to buy houses in up-and-coming neighborhoods to rent to families no longer acts as a broker. Instead, the relational work she performs is personal: matching herself to sellers and renters.

B From individual to *corporate* relational work

The definition above applies naturally to describe individual brokers working with individual clients, such as a real estate agent, an insurance broker representative, or a matchmaker. Much of the work of brokerage, however, occurs at the corporate, organizational, or institutional level, where brokers are firms, and clients may also be.

To understand what happens when relational work is outsourced to corporations, we must first discuss how corporations perform relational work in the first place, for themselves as organizational social actors. Relational work scholarship initially focused on the intimate sphere and interpersonal relationships (see Bandelj 2020, for a review). But relational work also takes place in the economic sphere and at the organizational level. The relational work tradition has acknowledged the organizational question without fully resolving it. Bandelj (2012) noted that relational work between representatives of organizations solidifies into macrolevel structures, while Zelizer (2012, pp. 156–157) catalogued instances of relational work by firms in the field of labor relations. Bandelj's (2020) review of "scaling up" relational work surveyed studies operating at the organizational level, but they mainly focused on what people do with and to each other within organizations, not on the corporate form itself as a vehicle for relational work.

Firms must conduct relational work to manage contingent, fleeting ties and longer-term relationships with workers, consumers, suppliers, lenders, investors, bureaucrats, and so on. The properties of the corporate form matter to this work and make it distinct. The firm is not simply a site where interpersonal relational work occurs; nor is it a singular, scaled-up actor that performs relational work like an individual would.

Firms are groups of people in designated roles, following norms and practices, supported by procedures, interfaces, and infrastructures, and doing the everyday work of managing the firm's relationship with its environment "at the coalface" (Gray and Silbey 2014; Huising and Silbey 2018). The mining metaphor directs attention to the frontline of the organization, where people enact its dealings with the outside world and translate organizational imperatives into routine practice. Relational work, like everything the corporation does in the world, is enacted through the people, procedures, infrastructure, and algorithms working at the coalface. Because of its collective and distributed nature, it is never visible in its entirety to any of the people involved.

Most firms are also legally organized as corporations. The corporation is a political entity with emergent capacities for coordination, information processing, and strategic action that exceed the sum of its parts (Ciepley 2013; Farrell and Newman 2023). There is a fundamental asymmetry between corporate actors and natural persons (Coleman 1982, 1990). The corporation aggregates information across thousands or millions of interactions, while each natural person sees only their own. The corporation can persist indefinitely, while individual relationships are transient. The corporation coordinates strategically across its full portfolio of relationships, while each counterparty acts in isolation. As a result, when a corporate actor interacts with natural persons, the corporate actor has structural and informational advantages.

These asymmetries are not accidental byproducts of size: they are legally constituted features of the corporate form. The corporation is one of the core modules of the legal code of capital (Pistor 2019). Corporate law endows the firm with durability that outlasts any individual relationship, priority that ranks its claims above those of other parties, and universality that makes those claims enforceable against the world, so that it can become a legal technology for concentrating and protecting accumulated advantages and leveraging them for productive purposes. These are the conditions under which corporate relational work takes place: it is inherently asymmetric, because of the different scales and material and informational resources of corporate and natural persons.

C Outsourcing relational work to corporations: the corporate brokers

When a client outsources relational work to a corporate broker, the corporate broker performs that work through people and practices at the coalface, each working in some small way to define relational packages, perform appropriate matches, and shape information flows. The corporate form reorganizes vicarious relational work, distributes it across roles, and embeds it in routines and procedures. To orchestrate this, corporate brokers develop an organizational apparatus comprised of internal and client-facing roles, information architectures and databases, contractual templates, and interfaces.

Saying a brokerage firm does relational work on behalf of its clients means that in sales meetings, in contract negotiations, in the design of user interfaces, in the defaults embedded in terms of service, and so on, people relying on organizational procedures are working to define the features of its clients' relationships, what transactions are appropriate to them, and what media may govern the exchanges. Just like for its own relational work, the corporation has a capacity for coordination, information aggregation, and temporal persistence that makes the vicarious relational work of corporate brokerage qualitatively different from what any individual could accomplish alone.

The corporate broker's relation with its clients is structurally asymmetric. With human clients, corporate and natural persons bring fundamentally different legal, informational, and temporal architectures to the relationship. Even with corporate clients, however, the asymmetry remains: the broker coordinates vicarious relational work strategically and benefits from cross-market information, while the client encounters it piecemeal. The matching of relations, transactions, and media moves from a pure negotiation between principals with comparable standing, who learn from each transaction, toward an administered process in which the corporate broker sets the terms, explicitly or behind the scenes, and captures the information that each pairing generates.

The data infrastructure of the corporate broker includes data taxonomies, client categories, risk models, algorithmic sorting procedures, and so on. This infrastructure does not merely facilitate transactions but actively constitutes relational packages by determining what transactions are possible, and what categories they should fall into. Classification systems and information infrastructures, like that of the corporate broker, are not neutral descriptions of the world; they actively organize social relations (Bowker and Star 1999; Fourcade and Healy 2024). The corporate broker's information infrastructure, much like a corporate rating agency's credit-scoring system or a platform's matching algorithm, sorts counterparties into categories the broker defines and uses those categories to allocate opportunities according to the broker's objectives. The impact of the broker's information infrastructure is magnified because the corporate broker works at scale, while individuals cannot usually design categories that simultaneously structure millions of relationships.

D The Broker Advantage

Why do people and firms outsource their relational work to brokers? To have the opportunity to broker, a participant in a matching market must hold either a *structural* or an *informational* advantage. The structural advantage depends purely on network topology and should translate into *access*: the broker can reach counterparties that clients cannot. The informational advantage should translate into *assessment*: the broker can evaluate prospective matches better than its clients can. In other words, the broker's service is valuable because it helps the client match with counterparties that the client could not otherwise easily find *or* evaluate.

Assessment involves two components: evaluating the general quality of the goods or counterparties involved and evaluating the quality of the specific pairing or relational package being considered. These components also correspond to two possible channels for the broker's information advantage: *attributional*, about how parties' attributes translate into individual quality, and *relational*, about which counterparties complement each other. Both are self-reinforcing, as the broker accumulates the knowledge required for assessment through the relational work it performs, and that learning makes it better at the work. The broker observes outcomes from many pairings across many principals, while each principal observes only its own. This gives the broker both a cross-market view and higher data volume.

The broker's information work can be informal and internal, as with a real estate broker who knows the local market, has an intuition for the right asking price, and holds social ties to likely buyers. It can also be formal and bureaucratized, as with a corporate mortgage broker that follows tables and formulas to determine which interest rate may produce successful matches between borrowers and lenders. The latter records and retains what the former holds only in personal experience, and this difference reflects that between individual and corporate brokerage.

Important models in the economics literature have focused on the attributional channel and characterized the broker's role primarily as quality certification (Li 1998) or expert screening (Bethune, Sultanum, and Trachter 2022): the broker identifies which goods or counterparties are high quality.

In that view, the broker is an appraiser whose cross-market experience helps it assess the general quality of counterparties more accurately than individual principals can. The relational-work view of brokerage suggests that the broker's value lies also, and often primarily, in understanding complementarities between counterparties and shaping relational packages accordingly. The broker is a relational worker and a matchmaker whose advantage comes from the ability to make pairings and to know which pairings will likely succeed, an ability that improves with repeated matchmaking.

IV ABM Results: The Broker's Advantage in a Simulated Matching Market

A Model and methods

To formalize and illustrate the theory developed in this article, this section describes an ABM of brokerage in a general matching market and presents the results of exploratory computational simulations. Their purpose is not to prove or disprove the theory, but to operationalize it, explore the dynamics it entails while providing concrete illustrations to the reader, and generate falsifiable propositions that future empirical work can use as hypotheses.

The theory makes claims about trajectories and feedback among heterogeneous, boundedly informed actors interacting repeatedly, making the ABM a natural tool to make its mechanisms explicit and sharpen its propositions.

The model is domain-agnostic and represents any market in which a broker facilitates pairwise matches and accumulates cross-market data from doing so. All agents use heuristic decision rules, and no agent is perfectly rational. One period equals one full update of the simulation loop; each period could be taken to represent one calendar month, one week, or one day, for instance, depending on the pace of the market that one wants to represent.

Pseudo-code and complete specifications for the model are included as **Appendix A and B**, respectively, and the code base for the project, implemented in Julia, is publicly available at `https://github.com/m-laprise/brokerage-abm`. It includes the scripts needed to reproduce the experiments that generated the data analyzed in this section, extract every number mentioned, create the figures and tables, and generate the results section in its entirety.

Setup. Many principals and a single broker, connected through an undirected network, inhabit the market. Each period, a principal draws a demand for new pairwise matches. Principals seeking matches search on their own for counterparties (self-search channel) or outsource to the broker (broker channel). The decision is made by comparing a running satisfaction score for each channel; principals who have never used the broker rely on its reputation (average satisfaction of current clients). After each period, each agent exits with probability η and is replaced by a fresh entrant.

Nature of the matching problem (Appendix B §2). Each principal has a fixed type, a vector of observable characteristics. A realized match between two principals yields an output equal to a fixed function of the two parties' types plus noise. The signal is unknown to everyone and must be learned. It has two parts, combined by a mixing weight ρ. The first is a general-quality term computed as the average of the two parties' quality scores, which depend on their type alone: a high-quality agent is high-quality for everyone. The second is a complementarity term that depends on the interaction of the two types: even two high-quality agents can make a bad match. At $\rho = 1$ match value is pure general quality, at $\rho = 0$ it is pure complementarity.

Difficulty of the matching problem (Appendix B §2c). The parameter δ determines how difficult the complementarity component is to learn. A pair's smooth interaction, $\mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j$, is multiplied by $1 + \delta$ on one side of a sharp boundary in type-pair space and by $1 - \delta$ on the other. At $\delta = 0$ the boundary has no effect. As δ increases, crossing it changes match value more sharply, so a learner must recover both the smooth interaction and which side of the boundary a pair occupies. The parameter acts only on complementarity and has no effect at $\rho = 1$, where match value is pure general quality.

Data collection and learning (Appendix A §3, algorithms 2–4; Appendix B §3). Principals record their own match history, with the type of each partner and the realized output of the pairing, brokered or not. The broker records the matches it mediates, with the type of both parties and the realized outputs, across all clients. Principals and the broker each fit a small neural network to their data using gradient descent, to learn to predict match output. At every period for the broker, and every other period for the principals, agents train on observations from the recent past to update their prediction function. To respect the symmetry of the unordered pairs in its training data, the broker uses as input a symmetric feature vector built from the type vectors.

As a robustness experiment, I repeat the complete design with Ridge regression replacing both neural networks. Principals regress output on the candidate's type; the broker uses the same symmetric additive and bilinear pair features as in the main model. Histories, training window, observation cap, fitting cadence, and reporting seeds are unchanged. Calibration on 10 baseline seeds excluded from reporting selected $\lambda_a = 0.003$ for principals and $\lambda_b = 0.001$ for the broker. This tests whether neural-network flexibility is necessary for the broker's ranking advantage. It does not yet distinguish whether that advantage comes from how much data the broker has or what those data contain.

To identify what about the broker's data produces this advantage, I run three additional Ridge versions at the baseline and on the $\rho \times \delta$ design. One limits the broker to no more observations than a typical principal. Another retains information about only one randomly selected party from each past match. The third retains both parties but omits the interaction between their types. Principal learning and all other settings are unchanged.

Search and match formation (Appendix B §§6–7). Self-search draws on known network neighbors and a small random pool of strangers; brokered search, on the broker's standing roster and current clients. Predicted values rank potential candidates; ties are ordered uniformly at random, so a constant predictor still produces a random ranking. Offers are made to the top candidates. For known partners, principals use the mean of their past experiences with them as the predicted value. Offers above the reservation threshold are accepted. When a match is made, its output is realized and both parties form a direct tie. Ties persist until one of the agents exits.

Economics (Appendix B §4). A match's realized output is its per-period value to each party. All quantities (reservation threshold, fees, search costs) are in common monetary units and are calibrated relative to the average match output over many random pairs. Every agent shares a reservation threshold (or outside option): offers are made or accepted only when predicted outputs exceed it. Self-search has a cost for each requested position whether filled or not, whereas the broker's fee is charged only on successful placements; both are an equal amount. Satisfaction scores net out costs or fees.

These modeling choices are the formal counterpart of the theoretical claims made above. Decomposing match output into general-quality and complementarity separates the attributional channel from the relational one. A principal learns from its own matches and observes only its counterpart's type, while the broker pools mediated matches across clients and observes both parties' types in each match.

Measurement (Appendix B §8). Diagnostic measures are recorded every period. In this article, structural position is measured as the broker's betweenness centrality in the graph, that is, the share of shortest paths between principals that run through the broker (Brandes 2001; Freeman 1977), as the standard *whole-network* measure of brokerage potential. The results would be substantively similar if examining the ego-network measures of Burt's constraint and effective size (Borgatti 1997;

Burt 1992; Everett and Borgatti 2020); Supplementary Figures S1–S4 report those analyses. The access fraction is the share of brokered matches in which the counterparty was not already tied to the demander. Prediction quality is measured on deterministic random-partner holdout samples evaluated against noiseless true output. Holdout evaluation reproduces the model's candidate-ranking behavior, including random ordering of tied predictions. Spearman rank correlation measures whether predictions order possible matches correctly, while R^2 measures level accuracy; gaps subtract the principal value from the broker value. R^2 varies much more because its finite-holdout denominator is the variance of noiseless holdout outputs, so modest prediction errors can produce extreme negative values when that variance is small. This volatility does not affect model behavior or the conclusions based on ranking, matching, and realized output because R^2 is diagnostic only and never enters a decision rule.

Results. All results are computed from a parameter sweep of the simulated market. A regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes. In the baseline regime, $N = 1{,}000$ agents interact for $T = 500$ periods with matching-function parameters $\rho = 0.5$ and $\delta = 0.5$, turnover $\eta = 0.02$, reservation threshold $f_r = 0.6$, initial network degree $k_G = 6$, broker roster share $\alpha_R = 0.2$, and self-search pool size $n_{\text{strangers}} = 10$. The simulation retains all periods, but time-series displays begin at $t = 30$ to omit the initialization transient. This display cutoff does not change the simulation or the late-period estimands. Around this baseline, I vary one parameter at a time ($\rho \in \{0, 0.3, 0.5, 0.7, 0.85, 1\}$, $\eta \in \{0, 0.001, 0.01, 0.02, 0.03\}$, $N \in \{500, 1000, 1500\}$, $f_r \in \{0.4, 0.6, 0.9, 1.2\}$, $\delta \in \{0, 0.25, 0.5, 0.75, 1\}$, $k_G \in \{4, 12\}$, $\alpha_R \in \{0.1, 0.2, 0.4\}$, and $n_{\text{strangers}} \in \{0, 10, 50\}$) and I sweep eight two-parameter grids: the six pairwise combinations of $\{\rho, \eta, N, f_r\}$ and the $\rho \times \delta$ grid over their full one-at-a-time levels, plus a focused $\rho \times f_r$ refinement over $\rho \in \{0.7, 0.85, 1\}$ and $f_r \in \{0.9, 1.05, 1.2\}$. In the six general grids, $\rho \times \eta$ and $\rho \times N$ use $\rho \in \{0, 0.5, 0.85, 1\}$, $\rho \times f_r$ uses $\rho \in \{0, 0.5, 1\}$, and the remaining axes reuse their one-at-a-time levels. The design comprises 98 regimes. Repeated grid coordinates that resolve to the same configuration belong to one regime, including $\rho = 1$ coordinates where δ is inactive. There are 20 independent seeds generally and 50 at the baseline (1990 runs in total). Within each regime, outcomes are averaged over a fixed time window within each seed and then across seeds. Late means use the final 20 periods ($t \in [481, 500]$) for all reported levels unless noted otherwise; early means use $t \in [50, 70]$ to quantify within-run drift. Cross-regime summaries give each regime equal weight. Remaining baseline parameter values are given in Appendix B §9. Monte Carlo convergence was assessed by comparing estimates from the first 10 seeds with all 20 seeds generally, and the first 20 with all 50 seeds at the baseline, and by computing Student-t 95% intervals from seed-level late-window means. Across the 14 non-R^2 outcomes shown in the figures, median relative interval half-widths across conditions ranged from 0.4% to 9.8%; median relative nested-seed changes ranged from 0.1% to 3.1%. Relative precision uses the absolute full-sample estimate as its denominator; regimes whose 95% interval contains zero are assessed from their absolute intervals instead.

B The broker primarily provides assessment, not access

Across the tested regimes, the broker is usually the dominant channel in the market. The outsourcing rate as a share of demand is 0.98 at the baseline (late mean, up from an early mean of 0.83) and has a median of 0.96 across regimes. It exceeds one-half in 95 of 98 regimes; the remaining regimes all have no turnover. Yet the broker primarily assesses options within the existing network rather than providing access beyond it. The access fraction (the share of brokered placements between principals with no pre-existing tie) has a late mean of 0.11 across regimes, down from 0.21 early, and never exceeds 0.23. It declines over time in all 98 regimes, with paired 95% intervals below zero in all 98. Most brokered matches therefore connect principals already tied to each other. Figure 1 and Table 1 summarize these baseline dynamics and the main measures.

Widespread broker use is generally, but not always, accompanied by higher realized output. Mean

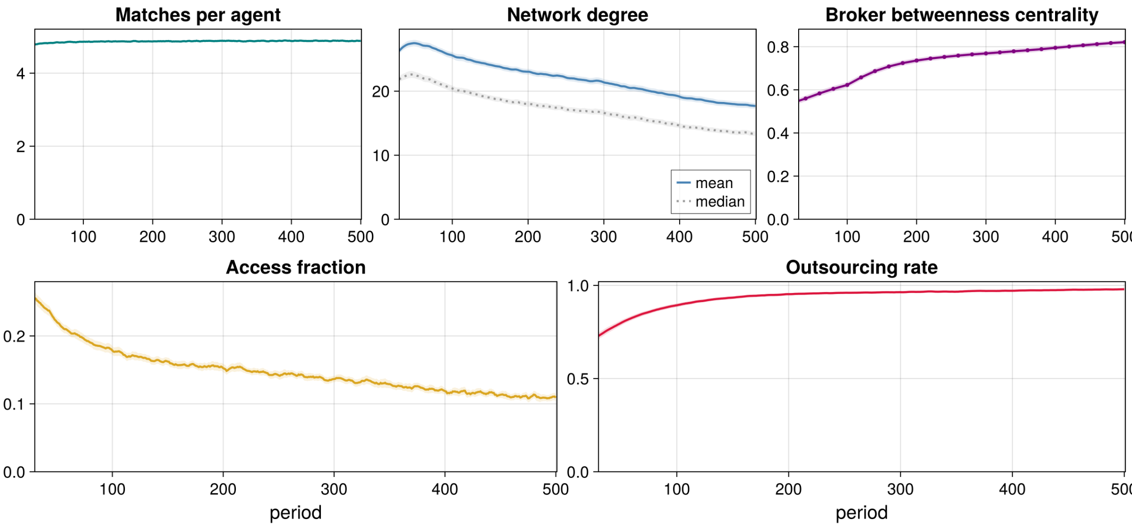

Figure 1: Baseline dynamics. Lines are 5-observation rolling ensemble means over seeds at the baseline regime; ribbons are pointwise 95% Monte Carlo intervals. Axes start at $t = 30$ to omit the initialization transient.

realized output is higher through the broker channel than through self-search in 84 regimes, and the output gap (broker minus self-search) averages +0.49 across regimes (late means). Paired 95% Monte Carlo intervals are above zero in 82 regimes and below zero in 14. The negative gaps occur in 14 regimes, all with no or nearly no turnover ($\eta \in \{0, 0.001\}$). Broker use and output differ because principals choose a channel using satisfaction scores that include unfilled requests, fees, and search costs, whereas the output gap compares realized output among formed matches. The gap also reflects the candidates considered, offers made, and offers mutually accepted. Outsourcing is therefore not itself evidence that the broker always produces better matches.

The broker's assessment advantage appears more consistently in ranking candidates than in predicting output levels. On a held-out sample of random partners (Appendix A §8), its rank correlation is 0.90 at the baseline and has a median of 0.90 across regimes, compared with 0.50 and 0.51 for principals. The broker-minus-principal rank gap is positive in 96 of 98 regimes. Paired 95% intervals are above zero in 92 regimes and below zero in only 1. The two nonpositive point estimates occur on the hardest pure-complementarity problems discussed below. The broker's level accuracy is less consistent, but not uniformly poor: its holdout R^2 is 0.63 at baseline, has a median of 0.40 across regimes, and is positive in 60 of 98 regimes. Candidate selection requires a useful ordering, not an accurate prediction of each candidate's absolute output.

Within the model, the broker's core service is choosing among accessible counterparties, not finding otherwise unreachable ones. Its ranking advantage appears under pure general quality as well as complementarity: at $\rho = 1$, the broker-minus-principal rank gap averages 0.46 and is positive in all 11 pure-quality regimes. The one-at-a-time checks also show that initial network degree (k_G), broker roster fraction, and the number of strangers available in self-search have comparatively small effects on the main economic outcomes.

C The broker's information advantage comes from pooled, pair-level data

The ordering advantage does not depend on neural-network learning. When Ridge regression replaces both neural networks, the broker-minus-principal rank gap is 0.335 at the baseline, has a median of

Table 1: Broker use, access, and assessment quality. Early and late are ensemble means over seeds averaged over $t \in [50, 70]$ and $t \in [481, 500]$; the across-regime column reports the median late mean across the 98 regimes. Outsourcing rate is outsourced requested positions divided by all requested positions.

	Baseline		Across regimes
Measure	early	late	late, median
Market use and access			
Outsourcing rate	0.83	0.98	0.96
Output gap (broker − self)	+0.02	+0.65	+0.52
Access fraction	0.21	0.11	0.11
Assessment quality			
Broker holdout rank correlation	0.84	0.90	0.90
Broker holdout R^2	0.23	0.63	0.40
Principal holdout rank correlation	0.55	0.50	0.51
Principal holdout R^2	−1.99	−3.16	−3.32

0.342 across the 98 regimes, and is positive in all 98. Supplementary Figure S5 reports the pattern across regimes and the $\rho \times \delta$ design. The paired-Ridge broker still has more observations and retains more information about each pair than a principal. Three additional Ridge experiments impose the restrictions described above. Because each restriction changes later matching and learning, their effects need not add up.

Pooled sample size accounts for much of the advantage (Fig. 2). Across the 26 regimes in the baseline and $\rho \times \delta$ design, the median rank gap is 0.458 under paired Ridge but only 0.065 with a size-matched broker. The size-matched gap remains positive in 20 of 26 regimes, so pooling is important without being the broker's only advantage.

The information retained in each observation also matters. When the broker retains the type of only one party, the median rank gap is -0.344 and is positive in only 6 regimes. When it retains both parties only through the sum of their types, omitting their interaction, the corresponding figures are -0.329 and 7. The cost of omitting the interaction is greatest when complementarity matters; under pure general quality, it is unnecessary. The broker's information advantage therefore comes from both the number of observations it pools and whether each observation retains information about both parties and their interaction, not from neural-network flexibility alone.

D The nature of the matching problem shapes the broker's advantage

Different markets present brokers with matching problems of different kinds. In this model, $\rho = 0$ gives pure complementarity and $\rho = 1$ gives pure general quality. The parameter δ increases learning difficulty by making complementarity payoffs change more sharply across a boundary in type-pair space. At $\rho = 1$, complementarity disappears and δ has no effect. This endpoint is therefore a separate boundary condition, not a smooth continuation of the interior ρ values. Broker centrality is governed primarily by the composition of match value (ρ). Betweenness centrality falls from 0.85 at $\rho = 0$ to 0.13 at $\rho = 1$, while the access fraction rises from 0.10 to 0.17 (late means along the one-at-a-time sweep, $\delta = 0.5$). The $\rho \times \eta$ grid shows the same relationship at every turnover rate: moving from $\rho = 0$ to $\rho = 0.85$ raises access and lowers centrality. When complementarity dominates, the broker is more central even though more of its matches re-pair principals who are already tied.

Difficulty changes the broker's informational advantage only when complementarity contributes substantially to match value. Under pure complementarity, increasing δ from 0 to 1 lowers the broker's rank correlation from 0.94 to 0.41 and changes the broker-minus-principal rank gap from

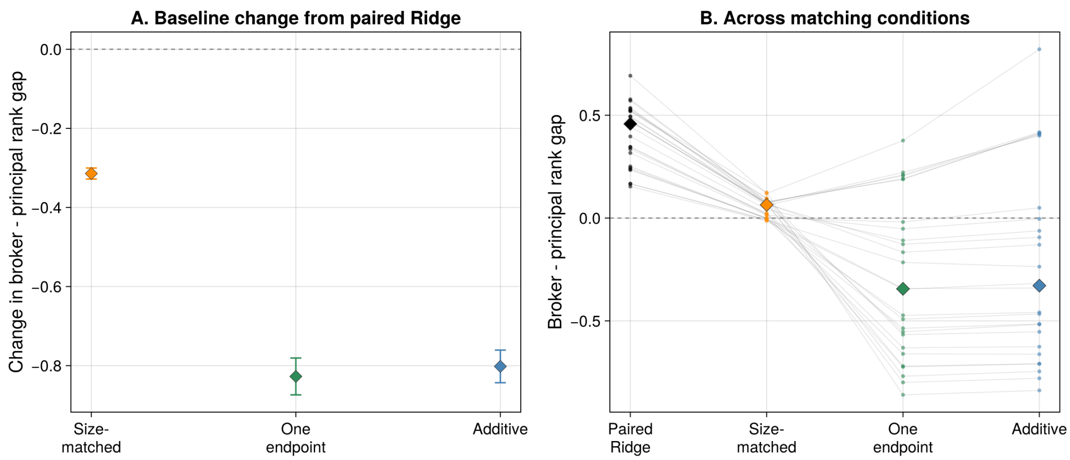

Figure 2: Sources of the broker's ranking advantage. Panel A shows each Ridge restriction's change in the broker-minus-principal rank gap relative to paired Ridge at the baseline. Points are mean common-seed differences; whiskers are paired 95% Monte Carlo intervals over 50 seeds. Panel B shows the absolute rank gap under paired Ridge and each restriction across the 26 regimes in the baseline and $\rho \times \delta$ design. Thin lines connect the same regime, small points are regime means, diamonds are medians, and the horizontal line marks zero.

0.28 to -0.04 (Fig. 3). This effect fades as ρ approaches general quality and disappears at $\rho = 1$ by construction. Paired 95% intervals for the rank gap are above zero in 24 of the 26 regimes. Only the two hardest pure-complementarity conditions depart from the general pattern: one interval includes zero and the other is below zero. This is a qualification to the neural-network result, not a general failure of broker learning. With paired Ridge, the rank gap remains positive in every regime.

The broker's realized output advantage is greatest when match value depends on complementarity. Paired 95% intervals for the broker-minus-self-search output gap are above zero in all 26 regimes, including the two hardest pure-complementarity conditions. This does not imply that poorer holdout rankings produce better matches. Holdout rank correlation evaluates random partner sets; realized channel output also reflects endogenous candidate pools, offers, and mutual acceptance. The broker's realized-output advantage survives where its relative NN ranking advantage does not, but the results do not identify which stage of the matching process accounts for that difference.

E A broker can be highly central while doing little bridging

In most regimes, the broker's structural position strengthens over time while the principal network thins. At the baseline, mean principal degree falls from 27.2 to 17.8, while broker betweenness centrality rises from 0.61 to 0.83 (early to late means). This co-movement occurs in 79 of 98 regimes, and principal degree falls in 80 of them (Fig. 4). In most regimes, direct ties do not accumulate fast enough to offset turnover. The broker's structural advantage therefore does not self-liquidate.

Assessment and access explain the composition pattern above. An access match creates a direct tie between previously unconnected principals and can remove the broker from their shortest path. An assessment match reuses an existing tie, so the broker can serve the market without directly densifying the principal network. As matching shifts toward general quality, brokered access becomes more common, the principal network becomes denser, and broker betweenness falls sharply even though outsourcing remains high. The broker's access work therefore creates alternative paths even while principals continue to rely heavily on its services. Figure 4 separates this composition pattern

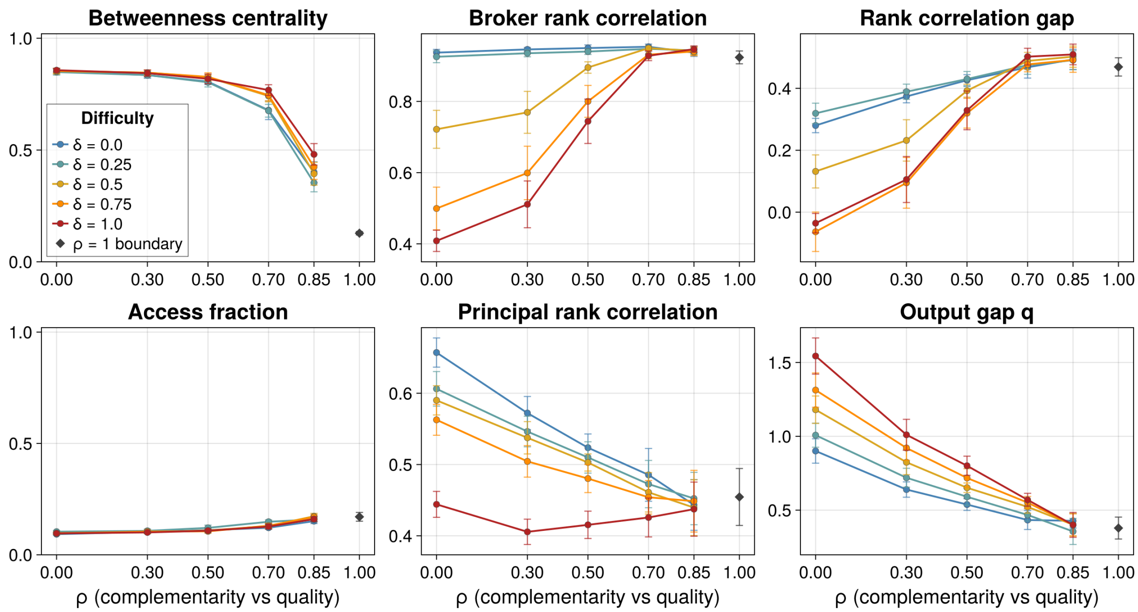

Figure 3: Outcomes across the matching problem ($\rho \times \delta$). Six outcomes across the $\rho \times \delta$ grid; one line per difficulty level δ, each spanning $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ has no effect, is one separate regime shown once and unconnected. The figure therefore displays all 26 regimes in the design. Each marker is a late-window mean over that regime's seeds; whiskers are 95% Monte Carlo intervals. The rank-correlation gap is broker minus principals; the output gap, broker channel minus self-search.

from turnover. Holding ρ fixed, increasing turnover raises access and centrality at every tested ρ. The access fraction therefore has no uniform association with centrality. Turnover creates network gaps; matching composition determines how often brokerage closes them with direct ties.

The turnover effect is substantial but nonlinear. Holding other parameters at baseline, increasing η from 0 to 0.03 lowers late mean principal degree from 85.6 to 16.6 and raises broker centrality from 0.11 to 0.84, with most of the change occurring between $\eta = 0.001$ and 0.01. Turnover also changes realized broker value. In the $\rho \times \eta$ grid, the 95% interval for the broker-minus-self-search output gap is below zero in 4 of 4 ρ conditions at $\eta = 0$, but above zero in 4 of 4 by $\eta = 0.01$. In the model, turnover supplies both structural opportunities and the conditions under which brokered matches outperform self-search.

The pure-quality boundary confirms that low access alone is not sufficient for high centrality. Although higher turnover raises centrality there, centrality remains below pure complementarity at the same turnover rate. In the tested regimes, high centrality occurs when turnover renews network gaps and brokerage mainly assesses existing relationships. Because access remains low overall and declines within runs, brokerage does little to eliminate the structural basis of its own position.

Figure 5 shows that structural position relates differently to relative ranking and realized value. Across otherwise matched parameter settings, moving from pure complementarity to pure general quality reduces the output gap from +0.87 to +0.23, while increasing the broker-minus-principal rank gap from 0.10 to 0.46. The broker is therefore most central and produces its largest realized-output advantage where its relative holdout-ranking advantage is smaller. Taken together, the results separate the sources of informational, economic, and structural advantage: pooled pair-level data improves ranking, complementarity raises broker value, and turnover renews the network gaps that sustain centrality. These advantages need not move together.

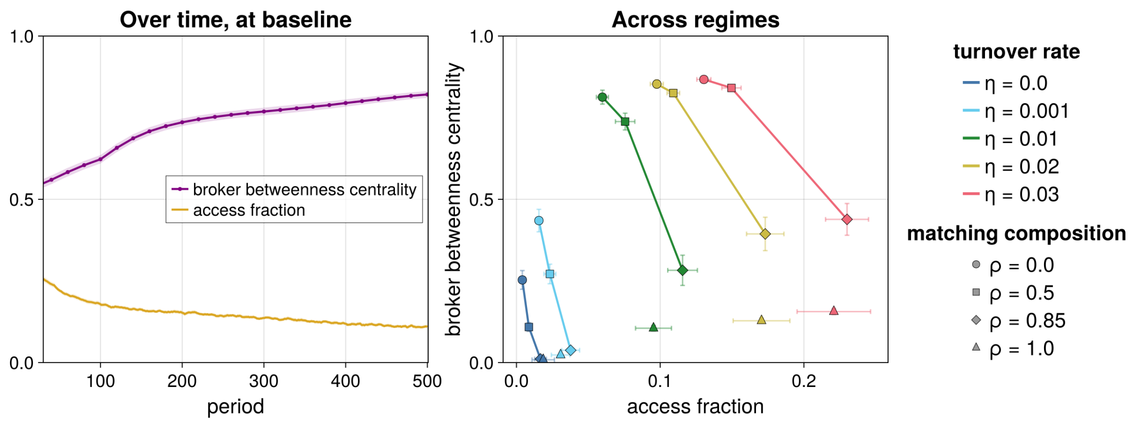

Figure 4: Broker centrality and access fraction. Left: broker betweenness centrality and access fraction over time at the baseline, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals; centrality is measured every 20 periods. Right: all 20 combinations in the $\rho \times \eta$ grid, with other parameters held at baseline, including $\delta = 0.5$, each shown as a late-window mean over that regime's seeds. Color identifies turnover η, marker shape identifies ρ, and whiskers are 95% Monte Carlo intervals on both axes. Within each η, lines connect $\rho = 0, 0.5, 0.85$. The pure-quality boundary $\rho = 1$, where complementarity is absent, is shown unconnected. Time-series panels begin at $t = 30$ to omit the initialization transient.

F Propositions

The simulation results can be condensed into the following falsifiable propositions, offered as hypotheses for future empirical work:

P1. In complex matching markets, brokers add value primarily by assessing complementarity among reachable counterparties, not by bridging disconnected parties.

P2. A broker's betweenness centrality is produced by outsourcing volume and client satisfaction, it tracks how much complementarity matters to match value, and assessment-heavy brokers tend to retain or gain centrality as networks evolve.

P3. Bridging position and bridging behavior are generally negatively related across markets: the most central brokers tend to do less bridging, except where turnover continually creates new network gaps for them to bridge.

P4. Broker predictive advantage tracks the complexity of learning the complementarities relevant to match value, and it can diverge from structural centrality.

V Transient brokerage and its conditions of possibility

Outsourced relational work performed at scale by corporate brokers creates the conditions under which the broker can transition from intermediary to principal. The broker dismantles the relational package previously negotiated with its clients and replaces it with a new configuration organized around the broker's ownership of the resources it used to intermediate (*resource capture*), or the monetization of the information generated through prior matching activity (*data enclosure*).

These configurations are two qualitatively different forms of capture, which can be pursued in parallel and along with standard brokering. In *resource capture*, the broker takes inventory risk and starts acting as a principal in the market it used to intermediate. Counterparties interact with the broker only, without creating direct ties between each other. As the broker expands its principal activity and

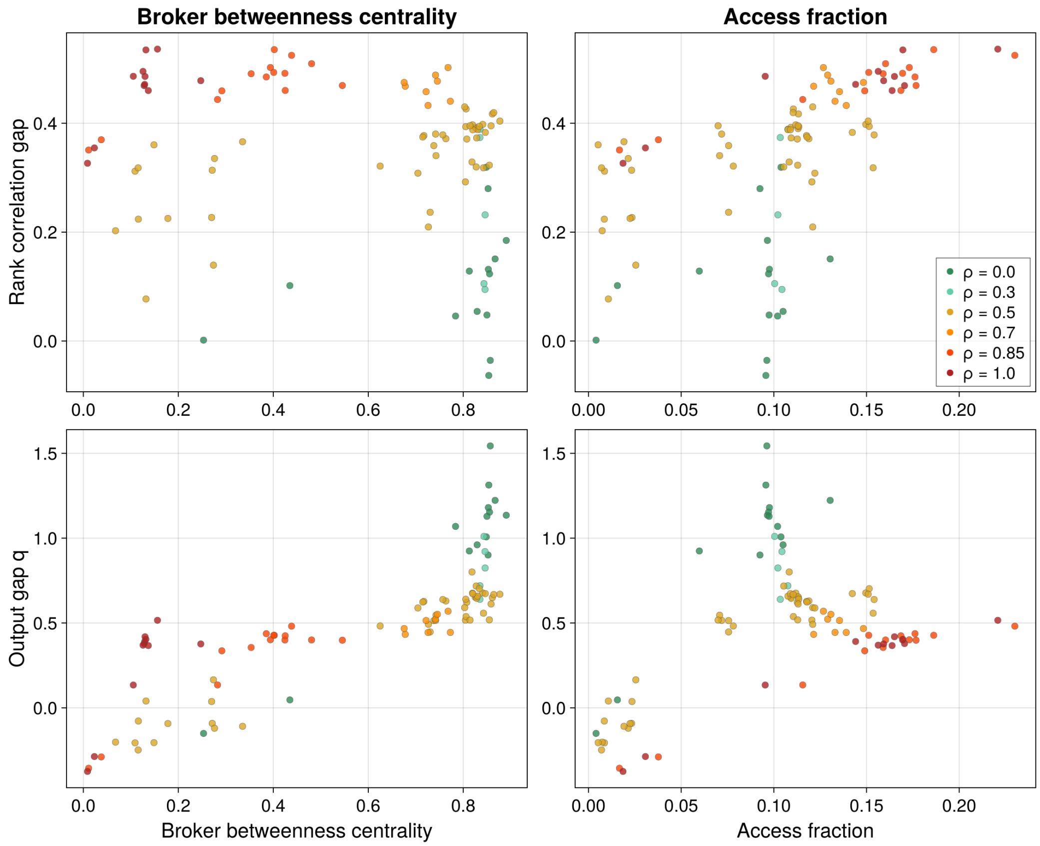

Figure 5: Prediction and output gaps vs centrality and access. Rank-correlation gap (broker minus principals) and output gap (broker channel minus self-search) against broker betweenness centrality and access fraction; one dot per regime, each a late-window mean over its seeds. Color identifies matching composition ρ.

shrinks its matching services, former clients become counterparties or competitors. In *data enclosure*, the broker monetizes information directly by selling predictions, analytics, or other information services, often as a subscription.

What makes capture possible, I suggest, is primarily the broker's information advantage, not its structural position. Structural and informational advantages are decoupled, and capture may occur even if the broker's centrality is declining. This is because constructing viable relational packages generates an informational byproduct that the broker can leverage. Two features of the broker's informational landscape, established through repeated matchmaking at scale, facilitate capture: cumulative knowledge of complementarities coming from the *outsourced* nature of its relational work, and information infrastructures arising from its *corporate* nature; both of which, other actors cannot easily replicate.

(i) The role of complementarity information. The broker's vicarious work generates knowledge of how different parties pair together, what kinds of matches succeed across the market, and what relational packages support different transactions. For regular market actors, this complementarity information is hard to replicate. Unlike attributional information about the general quality of counterparties, which any actor with sufficient market exposure can in principle acquire, complementarity information is generated through the repeated observation of various pairings. The broker holds it because it

does the matching for a variety of clients; clients, who experience only their own matches, do not—unless they purchase or otherwise acquire it. Complementarity information supports resource capture, because it allows the broker to identify which resources or counterparties are most likely to generate profitable matches, so that it can capture them strategically and offload them gainfully. Complementarity information also supports data capture because it can be packaged into prediction or analytics products and services.

(ii) The role of information infrastructures. The corporate broker's information infrastructure explains *how* its information advantage can be preserved and converted. The taxonomies, classification systems, contractual templates, and algorithmic sorting procedures that the broker develops to perform vicarious work at scale and in a distributed manner, through the everyday work of its employees and representatives, are also the infrastructures through which capture can be realized. They can be repurposed to route resources to the broker itself, to put these resources back in circulation profitably, or to organize data into a marketable product. Moreover, the corporate form gives the broker the legal privileges required to build, own, and protect these infrastructures over time (Coleman 1982; Pistor 2019).

These features, and transient brokerage itself, can be found in a wide variety of real-life markets. To make things more concrete, I provide two examples below, one for each configuration.

Dealers in collectible markets. Collectors of art, fine wine, rare books, or similar specialty goods may seek trades through broker dealers who know the market. Each collector has distinct tastes and holdings; match quality depends on multidimensional complementarity between what one party has and what another wants. Dealers accumulate knowledge of collector preferences across transactions, and they may decide to leverage it and start holding inventory, for instance, by launching and operating a gallery.

Interdealer brokerage in over-the-counter (OTC) financial markets. Interdealer brokers (IDBs) are specialized intermediaries in OTC financial markets (e.g. bond markets, foreign exchange, commodities). IDBs help institutions like banks and dealers conclude large, complicated, or anonymous transactions in the absence of a centralized trading place. They solve a hard matching problem: the right counterparty for a trade depends on inventory positions and risk appetites that parties wish to keep secret. IDBs must know who wants what at any given moment, they negotiate price across hidden order interest, they preserve anonymity until execution, and they manage long-term relationships with clients to remain included in the order flow. The complementarity information they accumulate, about who will absorb what positions at what price under what conditions, is substantial. Resource capture, however, faces regulatory and reputational hurdles. Instead, IDBs have monetized accumulated information through data divisions that package historical OTC pricing and other data products, and sell them to the same dealers whose trades generated the data and to a wider set of buy-side firms and regulators (see e.g. Parameta Solutions 2026; TP ICAP 2026).

In summary, three elements enable capture or enclosure: a matching problem that makes complementarity information valuable, an informational advantage accumulated through repeated vicarious relational work, and an institutional architecture that allows that information to be aggregated, organized, and protected.

VI Discussion

A Structural advantage as a product of relational work

Within the agent-based model, the broker's centrality is a product of its relational work. The broker is central because many principals outsource to it repeatedly, which they do because its assessments are good. Its centrality rises and falls with outsourcing and satisfaction, not with exogenous features of the initial graph.

With the production of structure, two patterns emerge that pre-existing theory may not have predicted. The first is that the broker tends to be most central exactly where it bridges least in practice. Across the regimes examined, centrality is negatively associated with the access fraction (Figure 4): in the complementarity-dominated markets where the broker is most prominent, almost all its matches re-pair principals already tied to one another. A strong bridging position coincides with weak bridging behavior. Occupying a strategic position and using it for one's work are different things that theory should not conflate.

The second pattern is about the sustainability of brokerage. A long-standing expectation in the tradition is that brokerage tends to be self-undermining (Burt 1992, 2002; Stovel, Golub, and Milgrom 2011). Each successful introduction densifies the network and dissolves the structural hole that gave the broker its advantage. Where the work is assessment, however, it leaves the network no denser than it found it. Across the simulations, access work is low and declining over time, so the broker's structural advantage tends to persist. Self-liquidation is a feature of access brokerage. Meanwhile, the assessment-heavy brokerage characteristic of complementarity-based matching markets is more durable.

The results also suggest a precise mechanism for the decoupling of structural and informational advantage posited in Section V. The two answer to different features of the matching problem. The broker's structural prominence tracks the composition of match value, rising as matching becomes more complementarity-driven, while its prediction edge tracks the complexity of the complementarity component, which can vary independently. A broker can be highly central despite a modest informational lead or hold a decisive informational lead from an unremarkable position. Betweenness centrality is best understood not as a static measure of opportunity but as a product of the relational work the broker performs.

B Scope conditions and generalizability

The concept of transient brokerage applies most directly to matching markets in which (i) matching is complex enough that the value of a match depends not only on the general quality of each party but also on their complementarity and (ii) a corporate broker operates as a repeat-player accumulating information across transactions. The further a case departs from these conditions, the weaker the predictions of the theory. When the relational dimension is weak, the broker's advantage rests primarily on data volume about multiple counterparties, making its informational position less privileged. When the broker is a physical person, she lacks the temporal persistence, infrastructural capacity, and legal architecture that the corporate broker relies on.

The dynamics of transient brokerage also depend on the nature of the brokered resource and the institutional environment. In a goods market where items are bulky and physical delivery is required, the costs and logistics of capture look very different, compared to a market for services, or a market where goods are virtual or ownership passes without physical possession. In some markets, regulation may slow or prevent capture. Market structure matters as well. The model considered in this article features a single broker. With multiple brokers competing for the same clients or matches, the cross-market data pool is fragmented, and each broker may accumulate a smaller informational advantage, unless one or a few dominate the others. A systematic treatment of broker competition is deferred to future work.

Transient brokerage is relevant to a particular configuration of brokerage, in which matching problem, corporate form, and information accumulation give rise to a distinctive trajectory. The sociological literature has used brokerage to describe a wide range of phenomena, from immigrant children translating for their parents to marriage matchmakers and political brokers. The relational-work view of brokerage developed in this article can be applied to what many of these so-called brokers do.

Without the repeat-player structure of corporate brokerage, the legal and infrastructural architecture of the firm, and an economic matching problem complex enough to sustain an informational advantage, however, the pathway to capture is not the same.

These scope conditions point to candidate domains in which to validate the theoretical propositions. Real estate and various types of financial intermediation, including interdealer brokerage, combine complex matching, corporate repeat-player brokers, and meaningful complementarity information. Import-export management and data markets also share many of these features.

C The political economy of corporate intermediation

The corporate broker is a vehicle for the accumulation of information generated by the interactions between others, and the principals whose matching activity produced the information do not participate in the profits. The asymmetries that make capture possible are not just a byproduct of a broker's size or expertise. They are legally constituted. Corporate brokers assemble cross-market data and build classification systems and information infrastructures under the legal protections of the corporate form, which allows vicarious relational work to solidify into a durable informational moat. Whether a broker can convert its informational position into resource ownership or data provision depends on the institutional environment as much as on network position.

The simulations make the distributional stakes of transient brokerage concrete. When the broker becomes a principal, it changes who gets connected to whom. The principals observe many fewer matches than they otherwise would, and most matches in the market teach them nothing at all. The informational asymmetry that makes capture possible is widened by capture itself: the broker takes control not only of the resource but of the data its placement generates. Transient brokerage shifts how value is distributed, channeling the returns of participants' collective relational activity to the corporations who record and mine the data it generates.

Capture by corporate brokers has become more common since the 1980s alongside three transformations that have reshaped market intermediation: the turn to finance, which expanded the share of economic activity routed through specialized intermediaries (Carruthers 2015; Krippner 2011); the digital turn, which dramatically lowered the cost of accumulating, storing, and analyzing the data that brokerage generates while also expanding the demand for predictive analytics (Cohen 2019; Gandy 1993); and the rise of platform capitalism, which produced multi-sided market operators that own the relational infrastructure of matching while monetizing the data it generates (Sadowski 2020; Salomé Viljoen, Goldenfein, and McGuigan 2021). Transient brokerage is one of the mechanisms that link these transformations together and contributes to a broader shift in how value is extracted from economic activity.

Significant evidence exists in the literature that, across financial markets, corporate brokers that once facilitated matches between others have, indeed, increasingly become principals, either by capturing the activity they used to intermediate or by monetizing data. In financial markets, the demutualization of stock exchanges is a prime example. As late as the mid-1990s, almost all major global exchanges operated as member-owned mutual organizations. By the mid-2000s, a majority had introduced electronic trading systems that allowed them to grant trading access directly to investors, bypassing the services of member firms, and had converted to for-profit shareholder structures (Aggarwal 2002; Petry 2021). For-profit exchanges now derive a growing share of revenue from proprietary data feeds and trading technology (MacKenzie 2021; Pardo-Guerra 2019).

A parallel transformation has taken place in interdealer brokerage for over-the-counter (OTC) markets, where the data divisions of prominent brokers now generate substantially higher operating margins than the broking activities themselves; this is the case, for instance, of the Parameta Solutions data division of TP ICAP, the largest interdealer broker (TP ICAP 2026). In consumer markets, Visa offers another example. Reorganized in 1970 as a member-owned cooperative of issuing banks, Visa operated for decades as a non-profit utility that brokered payment authorizations between cardholders, merchants, and banks (Stearns 2011). Following its 2007 restructuring and 2008 IPO, Visa converted into a for-profit corporation that now derives substantial revenue from data, analytics, and risk solutions services, alongside the network and processing fees generated by its core authorization and clearing activities (Visa Inc. 2024).

VII Conclusion

The very conditions that make brokerage possible and valuable, such as the scarcity of information and the complexity of the matching problem, create opportunities for the broker—particularly the corporate broker—to take control of resources. At the transition point, the broker dismantles the relational package that was negotiated with its principals and lets the boundary between *vicarious* and *personal* relational work collapse, for its economic advantage. It abandons the service-tie configuration that existed and (re)negotiates a new relationship (with the same principals or new ones), one that is premised on the broker's direct ownership and control of formerly brokered resources or of the data streams they generated. In the latter case, the accumulated data can also be mined internally or licensed out to feed proprietary behavioral prediction, classification, and scoring products (Fourcade and Healy 2017, 2024) for use in markets distinct from those in which they originated (Salome Viljoen 2021; Zuboff 2019).

This article helps explain why the digital turn has facilitated broker capture at scale. Because information is central to brokerage, changes in the information ecosystem reshape the conditions under which transient brokerage becomes available, and digital technology has changed those conditions decisively. Relational infrastructures that once consisted of internal handbooks, contractual templates, and bureaucratic procedures now incorporate data taxonomies, algorithmic sorting, and interface designs that constitute the relational packages they administer in real time and at scale. The asymmetries that make capture possible are correspondingly amplified, and the pathway from accumulated information to ownership or monetization, shorter.

A Digital platforms and automated matching

Transient brokerage is intimately related to the rise of digital platforms and algorithmic marketplaces, from gig economy apps to electronic exchanges, that partly automate the work of matching. Where brokers perform vicarious relational work on behalf of their clients, platform operators design mechanisms to automate and circumvent relational work, typically through auction protocols, ranking algorithms, and rating systems. The platform operator does not match buyer and seller; it builds the apparatus that matches them and extracts fees or data flows from the result.

In some markets, mechanism design substitutes for relational work. In others, it absorbs and reshapes relational work, with the platform operator administering the categories, ratings, and defaults that constitute the relational packages on behalf of both sides. These arrangements are not brokerage, but something different. The dynamics of their emergence share important features with transient brokerage: the platform operator accumulates cross-market information about its users, holds infrastructural and legal advantages that make this accumulation difficult to challenge, and is often well positioned to transition from automating matches to selling matched resources or analytics about them.

For example, Amazon operates as both a marketplace for third-party sellers and a direct retailer of its own products. Zhu and Liu (2018) tracked over 150,000 product listings initially supplied by third-party sellers and found that Amazon began directly selling roughly three percent of them over the following ten months. The platform was significantly more likely to enter listings with higher sales and better customer reviews, using information available to it as the marketplace operator. Khan

(2017, 2019) argues that platform operators in this position are best understood not as intermediaries but as integrated firms whose control of the matching infrastructure gives them a structural advantage over the sellers they host. The same configuration recurs beyond Amazon, across leading platform firms: their business model becomes that of a lean platform that owns and profits from the matching infrastructure while pushing the underlying productive activity and capital risk onto third parties (Rahman and Thelen 2019).

B Future Directions

Two main directions warrant future work. On the modeling side, the ABM developed here is exploratory. It has many interacting components and considerable researcher degrees of freedom in its specification, and it carries the usual caveats of this kind of hand-crafted simulation: its results illustrate the theory rather than establish it. Systematic investigation of the parameter space, the decision rules, and the model's sensitivity to them may surface dynamics that the results presented in this article do not reveal. The research capture and data enclosure mechanisms are introduced theoretically but not formalized; they are a natural next step for an extension. On the theoretical side, the framework does not yet fully address how platforms and algorithmic marketplaces reshape the dynamics of brokerage by automating the vicarious relational work that brokers historically performed. Both directions point toward the same underlying question about the impact of the digital turn on brokerage, the institutions that rely on it, and the actors that perform it or benefit from it.

The digital turn has brought about a series of transformations. The informational byproduct of relational matching has come to rival the value of the matches themselves. Relational work is increasingly performed for the data it generates and increasingly performed by machines rather than people. The informational asymmetry between brokers and their principals has widened. Existing relational packages are being dismantled to extract value from the information and infrastructures associated with them. These shifts redraw the boundaries on which brokerage rests: between internalized and outsourced relational work, and between vicarious and personal relational work. Brokerage becomes a transient position, held so long as brokering remains more profitable than capture.

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Appendix A
Simulation Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z, $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L. For a vector z, z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
Core		
N	scalar	Number of agents, with agent set $[N] = \{1, \ldots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t.
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \ldots; m^T]$	$\mathbb{R}^{T\times m}$	Measure table; row $m^t = \mathcal{M}_{t\cdot} \in \mathbb{R}^m$.
Parameters		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
L	learning function	NN (default) or Ridge.
λ_a, λ_b	positive scalars	Agent and broker Ridge slope penalties; calibrated experiment values are $\lambda_a = 0.003$ and $\lambda_b = 0.001$.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
E_{init}	scalar	Initial training steps.
E_{steps}	scalar	Minimum (floor) training steps per period.
η_{lr}	scalar	MLP Adam learning rate.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
Agent types, networks, and output		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N+1\}$; node $N+1$ is the broker.
$X^t = [x_1^t, \ldots, x_N^t]$	$\mathbb{R}^{d\times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0,1] \to \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j).
True matching function		

Symbol	Dimension/domain	Definition
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Matching difficulty: amplitude of the pair-dependent up/down multiplier.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d\times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d\times d}$	Symmetric regime operator.
Calibration and costs		
$\bar{q}_{\mathrm{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
Learning and predictions		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d+1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d\times n_i^t}$, $y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\cdot\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d\times n_b^t}$, $y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	predictor parameters	Agent i NN weights or Ridge coefficients.
Θ_b^t	predictor parameters	Broker NN weights or Ridge coefficients.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent NN optimizer moments and step counters; unused by Ridge.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i,j)$	scalar	Symmetric broker prediction for unordered pair $\{i,j\}$.
$w_{j\leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_{i\cdot}^t$	vector	Agent i's partner-indexed empirical means.
$C_{i\cdot}^t$	vector	Agent i's partner-indexed counts.
Period state and market objects		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i.
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \begin{pmatrix} \vartheta, \mathsf{rng}^t, \gamma, c, A, B, \bar{q}_{\text{cal}}, r, \phi, c_s, X^t, G^t, \\ \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_{i\cdot}^t, C_{i\cdot}^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \\ \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \end{pmatrix}.$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

The matching function is

$$f(x_i, x_j) = \rho \, \frac{x_i^\top c + x_j^\top c}{2} + (1-\rho)\, g(x_i, x_j)\, x_i^\top A x_j, \qquad g(x_i, x_j) = 1 + \delta \, \operatorname{sign}(x_i^\top B x_j).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.
Ensure: Final state S^T and measure table $\mathcal{M} \in \mathbb{R}^{T\times m}$.

1: $S^0 \leftarrow \textsc{Initialize}(\vartheta, \text{seed})$
2: **for** $t = 1, \dots, T$ **do**
3: $\quad (S^t, \mathcal{R}^t, Y^t) \leftarrow \textsc{PeriodUpdate}(S^{t-1}, t, \vartheta)$
4: $\quad \mathcal{M}_{t\cdot} \leftarrow Y^t$
5: **return** $(S^T, \mathcal{M})$

3 Learning and Prediction

Let $\mathsf{L} \in \{\text{NN}, \text{Ridge}\}$ select the learning function. The default is $\mathsf{L} = \text{NN}$. Both branches use the same histories, period window, and observation cap. Inputs remain on their raw DGP scale.

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h,

$$\text{NN}_\Theta(\xi) = w_2^\top \, \text{ReLU}(W_1 \xi + b_1) + b_2, \qquad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h\times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \qquad \psi_b(x_i, x_j) = \left[x_i + x_j;\ \operatorname{vech}\!\left(\frac{x_i x_j^\top + x_j x_i^\top}{2} \right) \right].$$

Agent networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
L	learning function	NN or Ridge.

Local variables for training (continued)

Symbol	Dimension/domain	Definition
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
Θ	predictor parameters	NN parameters (W_1, b_1, w_2, b_2) or Ridge parameters (α, β).
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}$, $\mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
λ	positive scalar	Learner-specific Ridge slope penalty; intercepts are not penalized.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the agent and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants (0.9, 0.999, 10^{-8}).
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t.
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker NN update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d \times \lvert L_i \rvert}$, $\mathbb{R}^{\lvert L_i \rvert}$	Agent i training view of $\mathcal{H}_i^t$.
$Z_{b,\text{tr}}, y_{b,\text{tr}}$	matrix, $\mathbb{R}^{\lvert L_b \rvert}$	Broker fitting view using the selected input map.
r_ℓ	$\{1, 2\}$	Endpoint retained once when broker observation ℓ is recorded for the single-principal ablation.

Algorithm 2 TrainNN

Require: Network parameters Θ, inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E, learning rate η_{lr}, persistent Adam state (m, v, τ).

for $e = 1, \ldots, E$ **do**
  $g \leftarrow \nabla_\Theta \mathcal{L}(\Theta), \quad \mathcal{L}(\Theta) = n^{-1} \sum_{\ell=1}^{n} \left(\text{NN}_\Theta(\Xi_{\text{tr},\cdot\ell}) - y_{\text{tr},\ell}\right)^2$
  $\tau \leftarrow \tau + 1$
  $m \leftarrow \beta_1 m + (1 - \beta_1) g, \quad v \leftarrow \beta_2 v + (1 - \beta_2)\, g \odot g$
  $\Theta \leftarrow \Theta - \eta_{\text{lr}} \dfrac{m/(1 - \beta_1^\tau)}{\sqrt{v/(1 - \beta_2^\tau)} + \epsilon}$
return $\Theta, (m, v, \tau)$

Algorithm 3 FitPredictor

Require: Learning function L, parameters Θ, raw inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, learner-specific Ridge penalty λ, and fitting parameters.

if $\mathsf{L} = \text{NN}$ **then**
  $E \leftarrow \max\{E_{\text{steps}}, \lceil E_{\text{init}} \Delta / n_{\text{total}} \rceil\}$.
  $(\Theta, \mathcal{O}) \leftarrow \textsc{TrainNN}(\Theta, \Xi_{\text{tr}}, y_{\text{tr}}, E, \eta_{\text{lr}}, \mathcal{O})$.
else

$(\alpha, \beta) \leftarrow \arg\min_{a,b} \left\{ n^{-1} \sum_{\ell=1}^{n} (a + b^\top \Xi_{\mathrm{tr},\cdot\ell} - y_{\mathrm{tr},\ell})^2 + \lambda \|b\|_2^2 \right\}.$

$\Theta \leftarrow (\alpha, \beta).$

return Θ and, for NN, $\mathcal{O}$.

The agent input is always the prospective partner type. The broker Ridge training input for observation ℓ is

$$z_{b\ell} = \begin{cases} \psi_b(x_{1\ell}, x_{2\ell}), & a_b \in \{\text{pair}, \text{size-matched}\}, \\ x_{r_\ell,\ell}, & a_b = \text{single-principal}, \\ x_{1\ell} + x_{2\ell}, & a_b = \text{additive}. \end{cases}$$

For size-matched history, uniformly sample without replacement from the broker's windowed and capped observations. Its sample size is the smaller of the broker sample size and the nearest-integer median agent sample size among agents with nonempty histories, with half-integers rounded up.

Prediction is modular:

$$\textsc{PredictAgent}(i, x_j) = \begin{cases} \mathrm{NN}_{\Theta_i}(x_j), & \mathsf{L} = \mathrm{NN}, \\ \alpha_i + \beta_i^\top x_j, & \mathsf{L} = \mathrm{Ridge}, \end{cases}$$

and

$$\textsc{PredictBroker}(i, j) = \begin{cases} \mathrm{NN}_{\Theta_b}(\psi_b(x_i, x_j)), & \mathsf{L} = \mathrm{NN}, \\ \alpha_b + \gamma_b^\top \psi_b(x_i, x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b \in \{\text{pair}, \text{size-matched}\}, \\ \alpha_b + \gamma_b^\top (x_i + x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{additive}, \\ g_b(x_i) + g_b(x_j) - \bar{y}_{b,\mathrm{tr}}, & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{single-principal}, \end{cases}$$

where $g_b(x) = \alpha_b + \gamma_b^\top x$. Before any Ridge fit, $\alpha = Q$, slopes are zero, and $\bar{y}_{b,\mathrm{tr}} = Q$.

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 4 TrainPredictors

Require: State S, period t, learning function L, and fitting parameters.

for $i = 1, \ldots, N$ in parallel **do**

  $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.

  if $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**

    Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \le W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \ldots, n_i\}$. If $|L_i| > M$, keep M spaced indices across L_i, including its first and last indices (or only its last index when $M = 1$).

    $\Xi_{i,\mathrm{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\mathrm{tr}} \leftarrow (y_i^t)_{L_i}$.

    $\Theta_i \leftarrow \textsc{FitPredictor}(\mathsf{L}, \Theta_i, \Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}, \lambda_a, \Delta_i, n_i, \vartheta)$; $\Delta_i \leftarrow 0$.

$n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.

if $n_b > 0$ and $\Delta_b > 0$ **then**

  Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \le W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \ldots, n_b\}$. If $|L_b| > M$, keep M spaced indices across L_b, including its first and last indices (or only its last index when $M = 1$).

  If $\mathsf{L} = \mathrm{Ridge}$ and $a_b =$ size-matched, replace L_b by the uniform size-matched sample defined above.

  Construct $Z_{b,\mathrm{tr}} = [z_{b\ell}]_{\ell \in L_b}$ and $y_{b,\mathrm{tr}} \leftarrow (y_b^t)_{L_b}$.

  $\Theta_b \leftarrow \textsc{FitPredictor}(\mathsf{L}, \Theta_b, Z_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}, \lambda_b, \Delta_b, n_b, \vartheta)$; $\Delta_b \leftarrow 0$.

---

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10{,}000} \sum_{\ell=1}^{10{,}000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$
$$r = f_r\, \bar{q}_{\text{cal}}, \qquad \phi = c_s = \lambda_c\, \bar{q}_{\text{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s.
f_r	scalar	Reservation coefficient; sets $r = f_r\, \bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ.
ψ_ℓ	scalar	Phase for active curve coordinate ℓ.
A_0	$\mathbb{R}^{d\times d}$	Raw matrix used to construct A.
Σ_x	$\mathbb{R}^{d\times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d\times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d\times d}$	Orthogonalized pre-normalization regime matrix.
$Z_{b,\text{seed}}, y_{b,\text{seed}}$	matrix, $\mathbb{R}^{n_b^0}$	Broker seed fitting batch using the selected input map.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i.
π^{ent}_{ij}	scalar	Entrant attachment probability.
$\mathcal{V}^{\text{ent}}_i$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 5 InitializeAgent

Require: State S, agent slot i, period counter t, parameters ϑ.
Ensure: Agent slot i is initialized in S.

Draw $u_i \sim \text{Unif}[0,1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
$x_i^t \leftarrow (\gamma(u_i) + \epsilon_i)/\|\gamma(u_i) + \epsilon_i\|_2$.
Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}^t_{i\cdot}$, $C^t_{i\cdot}$, χ^t_i, M^t_i, and Δ^t_i.
Initialize the selected predictor to the constant Q: for NN, draw W^t_{1i} with He initialization and set $b^t_{1i} = 0$, $w^t_{2i} = 0$, $b^t_{2i} = Q$, $\mathcal{O}^t_i = 0$; for Ridge, set $\alpha^t_i = Q$ and $\beta^t_i = 0$.
if $t = 0$ **then**
    $s^t_i \leftarrow (0, 0)$.
else
    Remove i from Roster^t, D^t, and M^t_j for all j; delete all incident edges in G^t; set $\bar{q}^t_{ji}, C^t_{ji} \leftarrow 0$ for all $j \neq i$.

Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

Add edges (i,j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.

$s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$

$s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.

Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.

Algorithm 6 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0.

Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1,\ldots,5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \ldots, d_\gamma$, defining γ.

Allocate empty agent and broker containers in S^0 and store γ.

for $i = 1, \ldots, N$ **do**

  InitializeAgent$(S^0, i, 0, \vartheta)$.

Draw $u_c \sim \text{Unif}[0,1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow \gamma(u_c) + \epsilon_c$ without re-normalizing.

Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \operatorname{tr}(A_0^\top A_0)$.

$\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.

Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\operatorname{tr}(\Sigma_x B_0 \Sigma_x A)}{\operatorname{tr}(\Sigma_x A \Sigma_x A)} A, \qquad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.

Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire}; add isolated broker node $N+1$.

Initialize $\mathcal{H}_b^0$ and the selected broker predictor to the constant Q using the corresponding NN or Ridge initialization above; set $\text{rep}_b^0 = 0$.

Sample $\text{Roster}^0 \subset [N]$ uniformly with $|\text{Roster}^0| = \lceil \alpha_R N \rceil$; add broker edges $(i, N+1)$ for all roster members.

for each unordered non-broker edge $\{i,j\} \in \mathcal{E}_G^0$ **do**

  Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.

Let $\mathcal{B}_{\text{seed}} \leftarrow \{\{i,j\} \in \mathcal{E}_G^0 : i,j \in \text{Roster}^0\}$; shuffle and keep at most 100 edges.

for each $\{i,j\} \in \mathcal{B}_{\text{seed}}$ **do**

  Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0.

  If $\mathsf{L} = \text{Ridge}$ and $a_b = \text{single-principal}$, draw and store $r_\ell \sim \text{Unif}\{1,2\}$ for this observation.

if $n_b^0 > 0$ **then**

  Set rep_b^0 to mean broker seed output.

else

  Set $\text{rep}_b^0 \leftarrow 0$.

Set $s_{i,\text{self}}^0$ to mean own seed output and $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$.

Set $\mathbf{p}_i^0 \leftarrow (n_i^0)$ for every agent and $\mathbf{p}_b^0 \leftarrow (n_b^0)$, so initialization is period 0; set $\Delta_i^0 \leftarrow n_i^0$ and $\Delta_b^0 \leftarrow n_b^0$.

for each i with $n_i^0 > 0$ **do**

$\Theta_i^0 \leftarrow \textsc{FitPredictor}(\mathsf{L}, \Theta_i^0, \Xi_i^0, y_i^0, \lambda_a, n_i^0, n_i^0, \vartheta)$; $\Delta_i^0 \leftarrow 0$.
if $n_b^0 > 0$ **then**
$Z_{b,\text{seed}} \leftarrow [z_{b\ell}]_{\ell=1}^{n_b^0}$, $y_{b,\text{seed}} \leftarrow y_b^0$.
$\Theta_b^0 \leftarrow \textsc{FitPredictor}(\mathsf{L}, \Theta_b^0, Z_{b,\text{seed}}, y_{b,\text{seed}}, \lambda_b, n_b^0, n_b^0, \vartheta)$; $\Delta_b^0 \leftarrow 0$.
return S^0.

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 7 ChooseChannel

Require: Agent i state (s_i, χ_i), broker reputation rep_b, RNG.
$U_{\text{self}} \leftarrow s_{i,\text{self}}$.
$U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1-\chi_i)\text{rep}_b$.
if $U_{\text{self}} > U_{\text{broker}}$ **then**
  return self.
if $U_{\text{broker}} > U_{\text{self}}$ **then**
  return broker.
return self w.p. 1/2, else broker ▷ $U_{\text{self}} = U_{\text{broker}}$.

6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 8 PeriodUpdate

Require: Previous state S^{t-1}, period t, parameters ϑ.
Ensure: Updated state S^t, accepted records $\mathcal{R}^t$, and pre-turnover measure row Y^t.
Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i, and set $D^t \leftarrow \emptyset$.
Refresh roster: remove each roster member with probability p_{roster}, then replenish uniformly to size $\lceil \alpha_R N \rceil$.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
$\mathcal{D}^t \leftarrow \emptyset$.
for $i = 1, \ldots, N$ **do**

Draw $d_i^t \sim \mathrm{Binomial}(K, p_{\mathrm{demand}})$.
if $d_i^t > 0$ **then**
$z_i^t \leftarrow \textsc{ChooseChannel}(i, \mathrm{rep}_b^{t-1})$.
Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
if $z_i^t = \mathrm{broker}$ **then**
Add i to D^t.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \mathrm{Roster}^t \cup D^t\}$.
$\textsc{TrainPredictors}(S, t, \vartheta)$.
$\mathcal{R}^t \leftarrow \textsc{OfferMarket}(S, \mathcal{D}^t, \vartheta)$.
Synchronize broker-node edges to Roster^t, D^t, and current broker-channel matches.
$\textsc{UpdateSatisfaction}(S, \mathcal{D}^t, \mathcal{R}^t)$.
if $D^t \neq \emptyset$ **then**
$\mathrm{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i,\mathrm{broker}}^t$.
else
$\mathrm{rep}_b^t \leftarrow \mathrm{rep}_b^{t-1}$.
if $t \bmod \Delta_{\mathrm{net}} = 0$ **then**
Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph G^t.
$Y^t \leftarrow \textsc{CollectMeasures}(S^t, \mathcal{R}^t, \vartheta)$ on the post-matching, pre-turnover state.
Append n_i^t to $\mathbf{p}_i^t$ for every agent and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
for $i = 1, \ldots, N$ **do**
if agent i exits with probability η_{exit} **then**
$\textsc{InitializeAgent}(S^t, i, t, \vartheta)$.
Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
return $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \to \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \to \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i.
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i.
$n_{\mathrm{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i \to j}^{\mathrm{self},t}$	scalar	Self-offer score.
Ω_i	integer	Residual offer quota for demander i.
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \to j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t.
$\mathcal{U}_i^{\mathrm{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\mathrm{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.

Local variables for OfferMarket (continued)

Symbol	Dimension/domain	Definition
$\mathcal{I}_{b,i}$	list	Accessible broker candidates for client i, sorted by v_b^t.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i\to j}^{\text{self}}, u_{b,i\to j}^{\text{local}}, u_{b,i\to j}^{\text{global}}$	$[0,1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and o.from.
$z^\star$	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0,1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i,j) = \textsc{PredictBroker}(i,j).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i,j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined for every graph neighbor and for the non-neighbors in the demander-specific stranger pool:

$$v_{i\to j}^{\text{self},t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i),\ C_{ij}^t > 0, \\ \textsc{PredictAgent}(i, x_j^t), & j \in \mathcal{N}_G^t(i),\ C_{ij}^t = 0, \\ \textsc{PredictAgent}(i, x_j^t), & j \in U_i^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j\leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i,j) \in \mathcal{E}_G^t,\ C_{ji}^t > 0, \\ \textsc{PredictAgent}(j, x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i. Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 9 AddOffer

Require: Offer book O^t, unordered-pair set $\mathcal{P}_O^t$, offer (i,j,z,v).

if O_{ij}^t is defined **then**
    return 0.
$O_{ij}^t \leftarrow (i,j,z,v)$.
if O_{ji}^t is undefined **then**
    $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i,j\}\}$.
return 1.

Algorithm 10 OfferMarket

Require: State S, demand records $\mathcal{D}^t$, parameters ϑ.
Ensure: Accepted records $\mathcal{R}^t$.

For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.

$O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
for each $(i, \mathrm{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
  Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\mathrm{strangers}}, N)$.
  $\mathcal{U}_i^{\mathrm{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
  Remove from $\mathcal{U}_i^{\mathrm{self}}$ the elements of $\{i\} \cup M_i^t$.
  Draw independent $u_{i\to j}^{\mathrm{self}} \sim \mathrm{Unif}(0,1)$ for each feasible j.
  $\mathcal{I}_i^{\mathrm{self}} \leftarrow \mathrm{argsort}_{j \in \mathcal{U}_i^{\mathrm{self}}:\ v_{i\to j}^{\mathrm{self},t} > r}(-v_{i\to j}^{\mathrm{self},t}, u_{i\to j}^{\mathrm{self}})$.
  for the first $\min(\Omega_i, |\mathcal{I}_i^{\mathrm{self}}|)$ agents j in $\mathcal{I}_i^{\mathrm{self}}$ **do**
    $\textsc{AddOffer}(O^t, \mathcal{P}_O^t, i, j, \mathrm{self}, v_{i\to j}^{\mathrm{self},t})$.
$\mathcal{A}_b^t \leftarrow \mathrm{Roster}^t \cup D^t$.
$D_+^t \leftarrow \{i : (i, \mathrm{broker}, d_i) \in \mathcal{D}^t,\ \Omega_i > 0\}$.
$\mathcal{P}_b^t \leftarrow \{\{i,j\} : i \neq j,\ (i \in D_+^t, j \in \mathcal{A}_b^t) \text{ or } (j \in D_+^t, i \in \mathcal{A}_b^t)\}$.
$\mathcal{B}_b^t \leftarrow \emptyset$.
for each $i \in D_+^t$ **do**
  Draw independent $u_{b,i\to j}^{\mathrm{local}} \sim \mathrm{Unif}(0,1)$ for each feasible j.
  $\mathcal{I}_{b,i} \leftarrow \mathrm{argsort}_{j \in \mathcal{A}_b^t : \{i,j\} \in \mathcal{P}_b^t,\ v_b^t(i,j) > r}(-v_b^t(i,j), u_{b,i\to j}^{\mathrm{local}})$.
  for the first $\min(\Omega_i, |\mathcal{I}_{b,i}|)$ agents j in $\mathcal{I}_{b,i}$ **do**
    Append $(i, j, v_b^t(i,j))$ to $\mathcal{B}_b^t$.
Draw a new independent $u_{b,i\to j}^{\mathrm{global}} \sim \mathrm{Unif}(0,1)$ for each $(i,j,v) \in \mathcal{B}_b^t$.
Sort $\mathcal{B}_b^t$ lexicographically by $(-v, u_{b,i\to j}^{\mathrm{global}})$.
for each $(i,j,v) \in \mathcal{B}_b^t$ **do**
  if $\Omega_i > 0$ **then**
    $\iota \leftarrow \textsc{AddOffer}(O^t, \mathcal{P}_O^t, i, j, \mathrm{broker}, v)$.
    $\Omega_i \leftarrow \Omega_i - \iota$.
for each $\{i,j\} \in \mathcal{P}_O^t$ **do**
  $o_{ij} \leftarrow O_{ij}^t$ if defined, else $\emptyset$; $o_{ji} \leftarrow O_{ji}^t$ if defined, else $\emptyset$.
  if $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
    $\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
  else if $o_{ij} \neq \emptyset$ and $w_{j\leftarrow i}^t > r$ **then**
    $\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
  else if $o_{ji} \neq \emptyset$ and $w_{i\leftarrow j}^t > r$ **then**
    $\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
return $\mathcal{R}^t$.

Algorithm 11 FinalizeMatch

Require: State S, accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o'.
if $j \in M_i^t$ **then**
  return .
Draw $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
Append columns x_j^t to Ξ_i and x_i^t to Ξ_j; append q_{ij}^t to y_i and y_j; increment n_i and n_j.
Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
Add edge (i,j) to G^t; add j to M_i^t and i to M_j^t.
if $z = \mathrm{broker}$ or ($o' \neq \emptyset$ and $o'.z = \mathrm{broker}$) **then**
  Append the two party types x_i^t and x_j^t to $X_{b,1}$ and $X_{b,2}$; append q_{ij}^t to y_b; increment n_b.
  If $\mathsf{L} = \mathrm{Ridge}$ and $a_b = \text{single-principal}$, draw and store $r_\ell \sim \mathrm{Unif}\{1,2\}$ for this observation.

$z^\star \leftarrow$ broker.
else
$z^\star \leftarrow$ self.
Append accepted record $(i, j, z^\star, q_{ij}^t, v, o, o')$ to $\mathcal{R}^t$.

---

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i.
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 12 UpdateSatisfaction

Require: State S, demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.
for each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
$Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i\}\, \zeta.q.$
$n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i,\ o.z = \text{broker}\}.$
if $z_i = \text{self}$ **then**
$\tilde{q}_i^t \leftarrow Y_i/d_i - c_s.$
else
$\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i;\ \chi_i^t \leftarrow 1.$
$s_{i,z_i}^t \leftarrow (1-\omega)s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t.$

CollectMeasures. CollectMeasures($S^t, \mathcal{R}^t, \vartheta$) is a pure measure-collection function called after matching and before entry/exit turnover. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, agent-node degree summaries excluding the broker node, and cached broker network measures. The measures themselves do not feed back into S^{t+1}. For rank correlation, predicted values first produce a strict ranking: equal predictions are ordered uniformly at random, independently for the agent and broker, while equal realized outputs retain average ranks. Spearman correlation is computed only after this predicted ranking is formed. The same rule is used for each holdout candidate set and for pooled accepted directed offers within each selected-offer channel.

Appendix B
Model Specifications for an ABM of Brokerage in Matching Markets

This document specifies an agent-based model of brokered matching.

The model is a discrete-time simulation of a matching market with *agents* and a *broker*. Agents seeking pairwise matches either search on their own or outsource to the broker. All economic quantities (match output, fees, surplus) are in the same monetary units. All agents use heuristic decision rules. No agent holds beliefs about other agents' strategies.

The model specification covers agents (§1), the matching problem (§2), how agents learn to predict match quality (§3), match economics (§4), network structure and agent turnover (§5), search and broker access (§6), the outsourcing decision (§7), and period measures (§8). Baseline parameter values and reproducibility are documented in Part II (§9).

Part I. Model Specification

1. Agents and Time

The model runs for T discrete periods (default 500). It has N agents (default 1000), a single broker, and an undirected network G over the agents and broker.

Each agent i is characterized by:

- **Type $\mathbf{x}_i \in \mathbb{R}^d$**: a fixed vector of observable characteristics, drawn as specified below and used by the matching function (§2).
- **Current-period matches M_i^t**: active bilateral relationships involving i in period t. Each counterparty can appear at most once in M_i^t; incoming accepted offers are not capped.
- **Active demand bound K**: the maximum number of active demands an agent can initiate in a period (default $K = 5$; §6).
- **Experience history $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}$**: (other party's type, realized match output) pairs from all matches involving i (§3a).
- **Satisfaction indices $s_{i,c}^t$**: one scalar per search channel $c \in \{\text{self}, \text{broker}\}$, used in the outsourcing decision (§7).

Agent types

Type vectors are generated near a smooth one-dimensional curve on the unit sphere in $\mathbb{R}^d$. The curve is parameterized by a position $u \in [0, 1]$:

$$\mathbf{x}(u) = \frac{\tilde{\mathbf{x}}(u)}{\|\tilde{\mathbf{x}}(u)\|}, \qquad \tilde{x}_\ell(u) = \begin{cases} \sin(2\pi\nu_\ell u + \psi_\ell) & \ell = 1, \ldots, d_\gamma \\ 0 & \ell = d_\gamma + 1, \ldots, d \end{cases}$$

where $\nu_\ell \sim \text{Unif}\{1, 2, 3, 4, 5\}$ are random integer frequencies, $\psi_\ell \sim \text{Unif}[0, 2\pi)$ are random phases, both drawn once per simulation, and $d_\gamma \leq d$ is the number of **active dimensions**. The remaining dimensions enter only through noise.

Each agent is drawn at a random position $u_i \sim \text{Unif}[0, 1]$ on the curve, then perturbed:

$$\mathbf{x}_i = \frac{\mathbf{x}(u_i) + \boldsymbol{\epsilon}_i}{\|\mathbf{x}(u_i) + \boldsymbol{\epsilon}_i\|}, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d}\mathbf{I}_d\right)$$

Noise is applied in all d dimensions, then the vector is re-normalized. The scale $\sigma_x/\sqrt{d}$ keeps the expected displacement from the curve approximately σ_x for any d.

d_γ controls type-space complexity by setting how many dimensions carry curve signal.

Broker

The broker is a permanent node in G and has no type vector. Its state is:

- **Experience history** $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}$: (party 1 type, party 2 type, realized match output) triples from mediated matches (§3c). The party labels have no directional meaning because the broker's pair features are symmetric.
- **Roster** Roster^t: the broker's standing access base (§6b).
- **Current clients** D^t: agents who outsource to the broker in period t (§6b).
- **Reputation** rep^t: the average broker satisfaction of current clients (§7).

2. The Matching Problem

Match output is symmetric and pair-level. The deterministic signal has three parts: agent general quality, pair interaction, and a pair-dependent multiplier applied to the interaction.

2a. Match quality

Let q_{ij} be the per-period value of the match between agents i and j, measured in monetary units:

$$q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

where $Q = 1.0$ is a constant offset, $\sigma_\varepsilon = 0.10$, and ε_{ij} is idiosyncratic match variation.

The deterministic signal $f : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is unknown to all agents and fixed for the duration of the simulation:

$$f(\mathbf{x}_i, \mathbf{x}_j) = \rho \cdot \frac{1}{2}\left[\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c}\right] + (1 - \rho) \cdot g(\mathbf{x}_i, \mathbf{x}_j) \cdot \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j$$

where $\rho \in [0, 1]$ mixes general quality (§2b) and interaction (§2c).

2b. Agent general quality

$\mathbf{c}$ is an ideal type vector. Each agent's general-quality score is $\mathbf{x}_i^\top \mathbf{c}$.

A pair contributes the average score $\frac{1}{2}(\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c})$ to f. The ideal type $\mathbf{c} \in \mathbb{R}^d$ is drawn at initialization as a perturbation of a random point on the agent type curve:

$$u_c \sim \mathrm{Unif}[0,1], \qquad \boldsymbol{\epsilon}_c \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d}\mathbf{I}_d\right), \qquad \mathbf{c} = \mathbf{x}(u_c) + \boldsymbol{\epsilon}_c.$$

Unlike agent types, $\mathbf{c}$ is not re-normalized after the perturbation.

2c. Match-specific interaction

The interaction term is $g(\mathbf{x}_i, \mathbf{x}_j)\, \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j$, where g is a pair-dependent multiplier.

$\mathbf{A}$ is a symmetric positive definite interaction matrix. Matrix $\mathbf{A}$ gives a smooth bilinear interaction geometry, while $\mathbf{B}$ divides pairs into two regions of the same type space. $\mathbf{A}$ and $\mathbf{B}$ are fixed for the duration of the simulation.

Base interaction. Draw $\mathbf{M}_A \in \mathbb{R}^{d\times d}$ with iid $\mathcal{N}(0,1)$ entries and set

$$\mathbf{A} = \mathbf{M}_A^\top \mathbf{M}_A \cdot \frac{d}{\mathrm{tr}(\mathbf{M}_A^\top \mathbf{M}_A)}.$$

Thus $\mathbf{A}$ is symmetric positive definite and $\mathrm{tr}(\mathbf{A}) = d$.

For an isotropic unit vector $\mathbf{x}$, $E[\mathbf{x}\mathbf{x}^\top] = \mathbf{I}/d$, so $E[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \mathrm{tr}(\mathbf{A})/d = 1$. This normalization fixes the average interaction scale.

Difficulty parameter. A second symmetric matrix $\mathbf{B} \in \mathbb{R}^{d\times d}$ defines

$$g(\mathbf{x}_i, \mathbf{x}_j) = 1 + \delta \cdot \mathrm{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)$$

where $\delta \in [0,1]$ (default 0.5).

The multiplier depends only on the sign of $\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j$. Positive gives $1+\delta$, negative gives $1-\delta$, and zero gives 1. At $\delta = 0$, the boundary between the two regions has no effect. As δ increases, crossing it changes the interaction more sharply, which makes the complementarity component harder to learn. Because $\mathbf{B}$ is symmetric, the multiplier is symmetric in i and j.

The regime matrix $\mathbf{B}$ is constructed so the regime split is not a rescaling of the base interaction on the realized type distribution. Let

$$\boldsymbol{\Sigma}_x = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i \mathbf{x}_i^\top$$

be the uncentered second-moment matrix of the realized type vectors.

To draw the raw regime matrix, draw $\mathbf{Z} \in \mathbb{R}^{d\times d}$ with iid $\mathcal{N}(0,1)$ entries, symmetrize it as $\mathbf{H} = (\mathbf{Z} + \mathbf{Z}^\top)/2$, and remove its average diagonal value:

$$\mathbf{B}_0 = \mathbf{H} - \frac{\mathrm{tr}(\mathbf{H})}{d}\mathbf{I}_d.$$

This makes $\mathbf{B}_0$ symmetric with zero trace.

Under the weighted inner product defined as

$$\langle \mathbf{M}, \mathbf{N} \rangle_{\boldsymbol{\Sigma}_x} = \operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{M} \boldsymbol{\Sigma}_x \mathbf{N}),$$

remove the projection of $\mathbf{B}_0$ onto $\mathbf{A}$ and normalize:

$$\mathbf{B}_{\text{raw}} = \mathbf{B}_0 - \frac{\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{B}_0 \boldsymbol{\Sigma}_x \mathbf{A})}{\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{A})} \mathbf{A}, \qquad \mathbf{B} = \frac{\mathbf{B}_{\text{raw}}}{\|\mathbf{B}_{\text{raw}}\|_F}.$$

This gives $\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{B} \boldsymbol{\Sigma}_x \mathbf{A}) = 0$ and $\|\mathbf{B}\|_F = 1$.

The weighted projection makes $\mathbf{B}$ orthogonal to $\mathbf{A}$ with respect to the realized type distribution, rather than with respect to unused directions in $\mathbb{R}^d$. In this context, orthogonality means

$$E\Big[(\mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j)(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)\Big] = \operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{B}) = 0,$$

where the expectation is over independent draws from the realized type distribution. Because agent types lie near a curve, this removes overlap between the two bilinear scores where agents actually occur.

$\mathbf{B}$ remains symmetric after projection and normalization.

Symmetry of $\mathbf{A}$ and $\mathbf{B}$ makes $f(\mathbf{x}_i, \mathbf{x}_j) = f(\mathbf{x}_j, \mathbf{x}_i)$.

3. Learning

Before searching for matches, agents and the broker may update their prediction models on their accumulated histories. Updated models are used to score candidates (§6). The broker is eligible to refit every period. Agent i is eligible when $i \bmod 2 = t \bmod 2$, so each agent is eligible every other period. An eligible model refits only when its history is nonempty and it has received new observations since its previous fitting call. A parameter switch selects the prediction model. Neural-network learning is the default. Ridge regression is the alternate experiment.

3a. Prediction models and fitting

In the default model, both agents and the broker use a fully-connected network with one hidden layer, ReLU activations, and a single linear output:

$$\hat{q}(\mathbf{z}) = \mathbf{w}_2^\top \operatorname{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + b_2$$

where $\mathbf{z}$ is the input feature vector and the remaining symbols are network weights and biases. Agents receive raw partner type vectors, the broker receives pairs of them.

Training rule. Network weights minimize MSE with the Adam optimizer (learning rate η_{lr}, default 0.01) on the current training window. No explicit weight regularization is applied. At $t = 0$, each network sets $\mathbf{W}_1$ using He initialization, $\mathbf{b}_1 = \mathbf{0}$, $\mathbf{w}_2 = \mathbf{0}$, and $b_2 = Q$ (§2a). An untrained predictor therefore returns exactly Q for every input. Networks with seed observations then take E_{init} gradient steps (default 200) on their seed histories.

Alternate Ridge rule. In the Ridge experiment, agents and the broker instead use

$$\hat{q}_i(\mathbf{x}_j) = \alpha_i + \boldsymbol{\beta}_i^\top \mathbf{x}_j, \qquad \hat{q}_b(\mathbf{x}_i, \mathbf{x}_j) = \alpha_b + \boldsymbol{\gamma}_b^\top \psi(\mathbf{x}_i, \mathbf{x}_j).$$

Agent and broker fits minimize mean squared error plus λ_a and λ_b, respectively, times the squared Euclidean norm of the slope coefficients. Intercepts are not penalized. Raw type and pair features are used without sample standardization. Before its first fit, a Ridge predictor has zero slopes and intercept Q. An initial joint baseline calibration applied the same seven candidate penalties from 10^{-4} to 1 to both learners over ten fixed seeds separate from the reporting seeds. Maximizing the median across seeds of the period 401–500 mean of the two holdout rank correlations selected a common penalty of 0.001, which we retain as λ_b. Holding λ_b fixed, a separate agent calibration compared $\lambda_a \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 0.5\}$ on the same seeds and selected $\lambda_a = 0.003$ by the median late-period agent holdout rank correlation.

Subsequent NN updates warm-start from the previous weights and persistent Adam moment state. The number of gradient steps adapts to the ratio of new observations to total history:

$$E_t = \max\left(E_{\text{steps}},\ \left\lceil E_{\text{init}} \cdot \frac{n_{\text{new}}}{n_{\text{total}}} \right\rceil\right)$$

where n_{new} is the number of observations added since that model's previous fitting call, $n_{\text{total}} = |\mathcal{H}^t|$ is full history size, and E_{steps} (default 100) is the minimum per-call step count. For an agent, n_{new} can span multiple periods because agents fit on an alternating schedule. Agents and the broker use the same NN step-count rule. Ridge is solved directly at each fitting call.

Initialization is period 0 for each learner. Its seed-history count is stored as the first period boundary. Each later boundary stores cumulative history at the end of a simulation period. A fitting call uses observations from the most recent $W_p = 40$ completed simulation periods. Before 40 simulation periods are available, the window also includes period 0. Once 40 are available, the period-0 seed history is excluded. If the selected history block exceeds $M_{\text{train}} = 2000$ observations, fitting deterministically keeps 2000 spaced observations across that block, including its oldest and newest observations. If the cap is one, it keeps the newest observation. The cap changes only the fitting sample, not the stored history or the period window. NN and Ridge use the same window, cap, fitting cadence, and histories.

3b. Agent i's model

History. $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}_{m=1}^{n_i}$ records the other party's type and realized match output from every match involving i.

Input. The agent's network takes the partner's type as input: $\mathbf{z} = \mathbf{x}_j$.

The hidden width is $h_A = 2d$, so the baseline uses 16 hidden units.

3c. Broker's model

History. $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}_{m=1}^{n_b}$ records both parties' types and realized match output from broker-mediated matches.

Input. The broker's network takes a symmetric pair feature vector:

$$\psi(\mathbf{x}_i, \mathbf{x}_j) = \left[\mathbf{x}_i + \mathbf{x}_j;\ \operatorname{vech}\left(\frac{\mathbf{x}_i\mathbf{x}_j^\top + \mathbf{x}_j\mathbf{x}_i^\top}{2}\right)\right].$$

The half-vectorization vech keeps the lower-triangular entries of the symmetric matrix, so the broker input dimension is $d_B = d + d(d+1)/2$, equal to 44 at $d = 8$.

The broker hidden width is $h_B = 8d$, so the baseline uses 64 hidden units.

Because $\psi(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_j, \mathbf{x}_i)$, each observed unordered pair contributes one training row.

The broker learns only from mediated matches. It does not observe self-search outcomes.

3d. Ridge broker ablations

Three separate Ridge-only ablations change the broker while leaving every agent's local Ridge model unchanged. **Size-matched pooled history** uniformly samples the broker's windowed and capped fitting data without replacement, using the smaller of its available sample size and the nearest-integer median sample size among agents with nonempty histories, with half-integers rounded up. **Single-principal history** randomly retains one endpoint when each broker observation is recorded, fits $g_b(\mathbf{x}) = \alpha_b + \boldsymbol{\gamma}_b^\top \mathbf{x}$, and scores a pair as $g_b(\mathbf{x}_i) + g_b(\mathbf{x}_j) - \bar{q}_b$, where $\bar{q}_b$ is the current fitting-sample output mean and defaults to Q before fitting. **Additive Ridge** uses $\mathbf{x}_i + \mathbf{x}_j$ without bilinear features. All use the calibrated agent and broker penalties, period window, observation cap, and fitting cadence.

4. Match Economics

Matches use a common reservation threshold and channel-specific cost accounting.

At initialization, 10,000 random agent pairs define the calibration mean $\bar{q}_{\text{cal}} = E[q]$. This calibration quantity scales r, ϕ, and c_s.

4a. Outside options

All agents share a reservation threshold r, calibrated at initialization:

$$r = f_r \cdot \bar{q}_{\text{cal}}$$

where $\bar{q}_{\text{cal}}$ is the calibration mean and f_r is the reservation coefficient (default 0.60). Since ϕ and c_s do not depend on r (§4c), f_r may take any nonnegative value, including $f_r \geq 1$, and is swept as an analysis axis.

4b. Participation constraints

Demanders emit offers only above the reservation threshold. Self-search uses the demander evaluation $\hat{q}_i(\mathbf{x}_j) > r$. Brokered search uses the broker pair prediction $\hat{q}_b(\{i,j\}) > r$. A one-sided offer is accepted only if the receiver's evaluation exceeds r; reciprocal offers are accepted automatically (§6d).

4c. Search costs

The self-search cost and broker fee share one calibrated level:

$$\phi = c_s = \lambda_c \cdot \bar{q}_{\text{cal}}.$$

λ_c is the shared cost rate (default 0.05). The two symbols differ in payoff accounting:

- **Self-search** costs c_s on each requested self-search position, whether or not it is filled.
- **Brokerage** charges ϕ only on successful brokered placements.

5. Network Structure and Turnover

Agents interact through a single undirected network G that determines search opportunities and structural position.

5a. Network initialization

G is initialized as a Watts-Strogatz small-world graph on the N agents (Watts & Strogatz, 1998), with ring degree $k_G = 6$, rewiring probability $p_{\text{rewire}} = 0.1$, and random agent order on the ring.

The broker is added as a permanent node with no type vector. Broker-node edges are synchronized to standing roster members, current-period broker clients, and agents currently engaged in broker-channel matches (§6b).

5b. Match tie formation

Each realized match adds an undirected i-j edge if one does not already exist. Ties persist until either node exits.

5c. Agent turnover

Agents exit independently each period with probability η_{exit} (default 0.02). An exiting agent is removed everywhere. Its node slot is reused by an entrant with a fresh type drawn from the same type process (§1). The entrant receives $\lfloor k_G/2 \rfloor$ edges to existing agents sampled with probability proportional to $\exp(-\|\mathbf{x}_{i'} - \mathbf{x}_j\|^2)$.

6. Search

At the start of each period, each agent draws active relationship demand

$$d_i \sim \text{Binomial}(K,\ p_{\text{demand}}),$$

where K is the maximum number of active demands the agent can initiate in the period and p_{demand} defaults to 0.50. If $d_i > 0$, the agent chooses **one channel for the batch** (§7): self-search or broker.

6a. Self-search

Agent i's self-search candidate pool has two components:

Neighbors. For a direct network neighbor j with prior pair-specific experience, agent i uses $\bar{q}_{ij} = \frac{1}{n_{ij}} \sum q_{ij}^{(m)}$, where n_{ij} is the number of times i and j have matched. If the graph edge exists but $n_{ij} = 0$, the agent instead uses $\hat{q}_i(\mathbf{x}_j)$. Initial histories seed one observation for every initial graph edge. Unobserved neighbor edges arise after turnover because entrants attach structurally before matching.

Strangers. Each self-searching demander i independently samples a pool U_i^t of $\min(n_{\text{strangers}}, N)$ agents uniformly without replacement, where $n_{\text{strangers}} = 10$ (default). The demander evaluates pool members that are not current neighbors using $\hat{q}_i(\mathbf{x}_j)$ (§3b).

Agent i orders feasible self-search candidates by these evaluations and emits up to d_i directed offers to candidates above r. Equal evaluations are ordered uniformly at random, so an agent whose predictions are constant produces a random ranking rather than an undefined ordering.

6b. Broker roster and access

The broker maintains a **roster** of agents it can propose as counterparties when mediating matches. In each period, it also observes the current-client set D^t of agents who outsourced. Broker access for the period is:

$$\mathcal{A}_b^t = \text{Roster}^t \cup D^t.$$

Current clients expand broker access for the period but do not become standing roster members.

Initialization. The roster is seeded with a target size

$$R^* = \lceil \alpha_R N \rceil,$$

where the broker roster share α_R defaults to 0.20. The model draws R^* agents uniformly at random (default 200 at $N = 1000$). The broker's history is seeded from existing roster-roster ties in the initial graph, capped at 100 observed ties.

Roster refresh. At the start of each period, each roster member independently exits with probability p_{roster} (default 0.02). If $\widetilde{\text{Roster}}^t$ is the post-churn roster,

$$\widetilde{\text{Roster}}^t = \{i \in \text{Roster}^{t-1} : u_i^t > p_{\text{roster}}\}, \qquad u_i^t \overset{iid}{\sim} \text{Unif}[0, 1],$$

then the broker samples without replacement from $\{1, \ldots, N\} \setminus \widetilde{\text{Roster}}^t$ until $|\text{Roster}^t| = R^*$, or until the population is exhausted. Broker-node edges are synchronized to the standing roster, current clients, and current broker-mediated matches (§5a).

6c. Broker-mediated search

For every unordered pair $\{i, j\}$ such that one side is a broker-client demander and the other side is in $\mathcal{A}_b^t$, the broker computes predicted match quality. For each broker-client demander i, it ranks accessible counterparties above r, breaking equal predictions uniformly at random, and keeps the highest-ranked candidates up to i's residual active broker demand. Directed broker offers are then processed in a global score ranking with a new, independent random ordering within score ties. If both sides are broker-client demanders with remaining active demand, the same unordered pair can generate reciprocal broker offers.

6d. Offer acceptance and match formation

All self-search and broker offers enter one shared market, grouped by unordered pair.

1. If pair $\{i, j\}$ contains reciprocal offers $i \to j$ and $j \to i$, the relationship is accepted automatically.
2. If pair $\{i, j\}$ contains one directed offer $i \to j$, receiver j evaluates i using $\bar{q}_{ji}$ for known partners and $\hat{q}_j(\mathbf{x}_i)$ otherwise. The offer is accepted iff this value exceeds r.
3. Each accepted unordered pair forms at most one current-period relationship.

At market finalization, for each accepted match: 1. Realized output is drawn: $q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}$. 2. Both parties add the observation to their histories: i adds $(\mathbf{x}_j, q_{ij})$ to $\mathcal{H}_i$; j adds $(\mathbf{x}_i, q_{ij})$ to $\mathcal{H}_j$. 3. If either directed offer used the broker channel, the broker adds $(\mathbf{x}_i, \mathbf{x}_j, q_{ij})$ to $\mathcal{H}_b$ once. 4. An edge is added between i and j in G (if not already present).

A demander can have at most d_i outgoing offers, but can receive any number of acceptable incoming offers. Thus K is an active-demand bound, not a capacity constraint on receiving offers.

Before the next period begins, match lists are cleared. The next period begins with no active relationships.

7. The Outsourcing Decision

7a. Satisfaction tracking

Each agent i maintains a satisfaction index $s^t_{i,c}$ for each search channel $c \in \{\text{self}, \text{broker}\}$. For the chosen channel, the index updates once per active-demand period as an EWMA of realized match value net of search costs:

$$s^{t+1}_{i,c} = (1 - \omega)\, s^t_{i,c} + \omega \cdot \tilde{q}$$

where $\omega = 0.2$ and $\tilde{q}$ is the period input in the table. The denominator is requested demand d_i, so unfilled requests contribute zero output. A reciprocal relationship can update two channel-specific states, one for each directed offer.

Channel	Satisfaction input $\tilde{q}$
Self-search	$\dfrac{\sum q_{ij}}{d_i} - c_s$, summing over accepted directed offers made by i through self-search
Brokered	$\dfrac{\sum q_{ij} - \phi \cdot n_{i,\text{broker success}}}{d_i}$, summing over accepted directed offers made by i through the broker

Brokered failure gives $\tilde{q} = 0$, self-search failure gives $\tilde{q} = -c_s$.

$s^t_{i,\text{self}}$ covers all self-search, including known neighbors.

At initialization, self-satisfaction is the mean of seed match outcomes and broker-satisfaction is the broker's seed-data reputation (§7c). Fresh entrants set self-satisfaction to mean neighbor self-satisfaction and broker-satisfaction to current broker reputation.

The first broker choice, regardless of placement success, switches the agent from broker reputation to its own $s^t_{i,\text{broker}}$ (§7b).

7b. Decision rule

Each agent with $d_i > 0$ compares:

- $\textbf{score}_{\textbf{self}} = s^t_{i,\text{self}}$
- $\textbf{score}_{\textbf{broker}} = s^t_{i,\text{broker}}$ if the agent has tried the broker, otherwise rep^t_b.

The agent chooses the higher-score channel. Ties are resolved by fair coin flip.

7c. Broker reputation

$$\text{rep}^{t+1}_b = \begin{cases} \frac{1}{|D^t|} \sum_{i \in D^t} s^{t+1}_{i,\text{broker}} & \text{if } D^t \neq \emptyset \\ \text{rep}^t_b & \text{otherwise} \end{cases}$$

where D^t is the set of agents who outsourced to the broker this period. With current clients, reputation is their mean post-update broker satisfaction; with no clients, it is unchanged. Reputation is initialized from the mean broker seed outcome.

8. Period Measures

Period measures are evaluated after current-period matching, satisfaction, and reputation updates, but before agent entry/exit turnover. Agent decisions do not depend on these measures.

Network measures

Broker network-position measures use G, including the broker node (§5a). They are computed when $t \bmod \Delta_{\text{net}} = 0$ (default 20). Other period measures are recorded each period.

Betweenness centrality. Freeman betweenness (Freeman, 1977) for the broker node:

$$\text{BC}_b = \frac{1}{(n-1)(n-2)} \sum_{u \neq b} \sum_{v \neq u} \frac{\sigma_{uv}(b)}{\sigma_{uv}}$$

where $n = N + 1$, σ_{uv} is the number of shortest paths from u to v, and $\sigma_{uv}(b)$ is the number passing through b. The double sum counts undirected pairs from both directions, so the denominator is $(n-1)(n-2)$ (Brandes, 2001, p. 9).

Burt's constraint. Broker ego-network constraint (Burt, 1992):

$$\text{Constraint}_b = \sum_j \left(p_{bj} + \sum_{h \neq b,j} p_{bh}\, p_{hj} \right)^2$$

where $p_{bj} = 1/d_b$ and $p_{hj} = 1/d_h$ (Everett & Borgatti, 2020). The indirect term uses intermediary h's degree.

Effective size. Non-redundant broker contacts using the Borgatti (1997) binary-undirected simplification:

$$\text{ES}_b = d_b - \frac{2t_b}{d_b},$$

where d_b is the broker's degree and t_b is the number of ties among the broker's neighbors, excluding ties to the broker.

Prediction quality

Prediction quality is evaluated on holdout pairs and accepted directed offers:

Sample	Construction	Reported measures
Holdout pairs	Up to 100 agents with non-empty histories; for each sampled agent i, $\min(40, N-1)$ non-self partners j. Agent and broker predictions are evaluated against noiseless $Q + f(\mathbf{x}_i, \mathbf{x}_j)$. Both models use the same deterministic sample conditional on seed and period.	Per-agent R^2, RMSE, bias, and Spearman rank correlation, averaged across sampled agents.
Selected offers	Accepted directed offers by channel, self-search or brokered. A reciprocal relationship can represent two directed decisions.	Channel-specific R^2, RMSE, bias, and Spearman rank correlation.

The definitions are

$$R^2 = 1 - \frac{\text{MSE}}{\text{Var}_{\text{pop}}(q)}, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum(\hat{q}-q)^2}, \qquad \text{Bias} = \frac{1}{n}\sum(\hat{q}-q).$$

Rank correlation evaluates the strict ranking produced from predicted output. Candidates are ordered by predicted output, with each set of equal predictions ordered uniformly at random. Agent and broker tie-breaking draws are independent. Spearman's ρ_S is then computed between this produced ranking and the realized-output ranks; equal realized outputs retain their average ranks. The same rule is applied to each holdout candidate set and to the pooled accepted directed offers within each selected-offer channel. Thus a constant predictor produces a random ranking with expected rank correlation zero. For selected-offer summaries, R^2, bias, and rank correlation are undefined when $n < 5$ or $\text{Var}_{\text{pop}}(q) < \sigma_\varepsilon^2/6$, while RMSE is defined whenever $n > 0$. Holdout summaries use common valid-agent support for all four measures: an agent-level holdout set that fails the sample-size or variance condition is excluded from the R^2, RMSE, bias, and rank-correlation means.

Other measures

Measure	Definition
Access vs. assessment	For each accepted broker-directed offer, record whether receiver j was a direct neighbor of sender i in G at the time of the offer. Non-neighbor receivers indicate access; neighbor receivers indicate assessment.
Match quality by channel	Average realized match output $\bar{q}_c^t$, where $c \in \{\text{self}, \text{brokered}\}$.
Mean channel satisfaction	Cross-agent means $N^{-1}\sum_i s_{i,\text{self}}^t$ and $N^{-1}\sum_i s_{i,\text{broker}}^t$.
Outsourcing rate	Outsourced requested positions divided by total requested positions.
Broker access size	$\|\mathcal{A}_b^t\| = \|\text{Roster}^t \cup D^t\|$, measured after outsourcing decisions form D^t and before entry/exit.
Counterparty concentration	Median and maximum current-period counterparties per agent.
Agent-node degree summaries	Mean, median, minimum, and maximum degree, excluding the broker node.

Part II. Baseline Values and Reproducibility

9. Baseline Values and Reproducibility

9a. Baseline parameter values

The table lists baseline values only. Entries are arranged in two columns; each column follows the document order, with the run length listed first.

The baseline learning function is NN. Ridge and its broker variants are alternate experiments (§3).

Symbol	Baseline	Defined in	Symbol	Baseline	Defined in
T	500	§1	f_r	0.60	§4a
N	1000	§1	λ_c	0.05	§4c
d	8	§1	k_G	6	§5a
d_γ	8	§1	p_{rewire}	0.1	§5a
σ_x	0.5	§1	η_{exit}	0.02	§5c
Q	1.0	§2a	K	5	§6
ρ	0.50	§2a	p_{demand}	0.50	§6
δ	0.5	§2c	$n_{\text{strangers}}$	10	§6a
σ_ε	0.10	§2a	α_R	0.20	§6b
η_{lr}	0.01	§3a	p_{roster}	0.02	§6b
E_{init}	200	§3a	ω	0.2	§7a
E_{steps}	100	§3a	Δ_{net}	20	§8
W_p	40 periods	§3a	M_{train}	2000	§3a

9b. Reproducibility

All model-event randomness flows from a single integer seed. The seed determines type draws, G, matching function parameters ($\mathbf{c}$, $\mathbf{A}$, $\mathbf{B}$), broker seed roster, standing-roster churn and replenishment, random ordering within market score ties, and subsequent model events. Holdout samples and their independent agent and broker tie-breaking draws are deterministic conditional on seed and period. Simulations are fully reproducible given parameters and seed.

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Freeman, L. C. (1977). A set of measures of centrality based on betweenness. *Sociometry*, *40*(1), 35–41.

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Supplementary Material

Alternative structural measures and Ridge robustness

The first four figures reproduce the main structural-advantage analyses with the broker's two other saved ego-network measures, Burt's aggregate network constraint and Burt's effective size. Both are recomputed for the broker every 20 periods. Figure S5 reports the focused paired-Ridge robustness analysis discussed in the main text. As in the main text, a regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes.

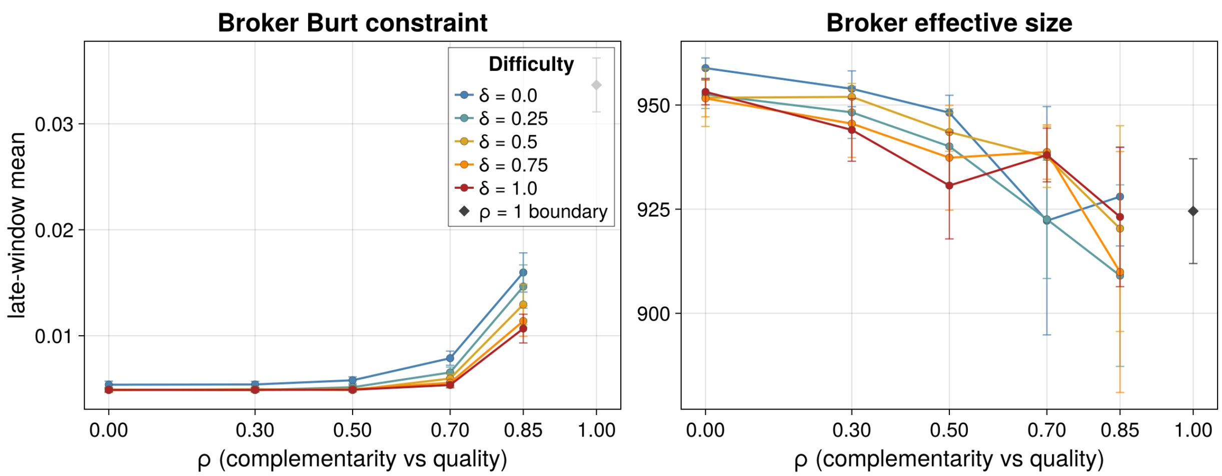

Figure S1: Broker Burt constraint (left) and broker effective size (right) across the $\rho \times \delta$ grid. Lines for each difficulty level δ span $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are late-window means over seeds and whiskers are 95% Monte Carlo intervals.

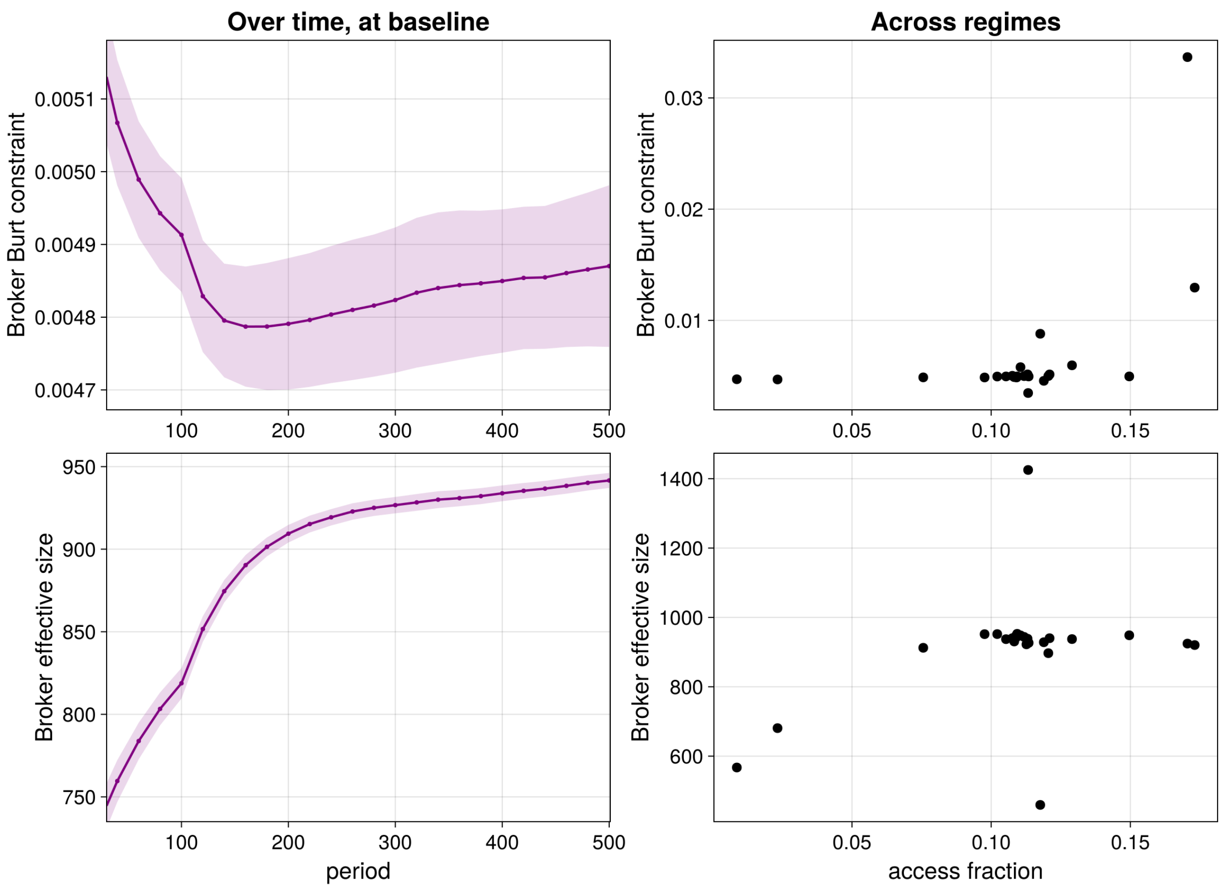

Figure S2: Broker Burt constraint (top) and broker effective size (bottom). Left: the measure over time at the baseline, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals, measured every 20 periods. Right: one dot per one-at-a-time regime, the measure against access fraction, each a late-window mean over that regime's seeds. Time starts at $t = 30$ to omit the initialization transient.

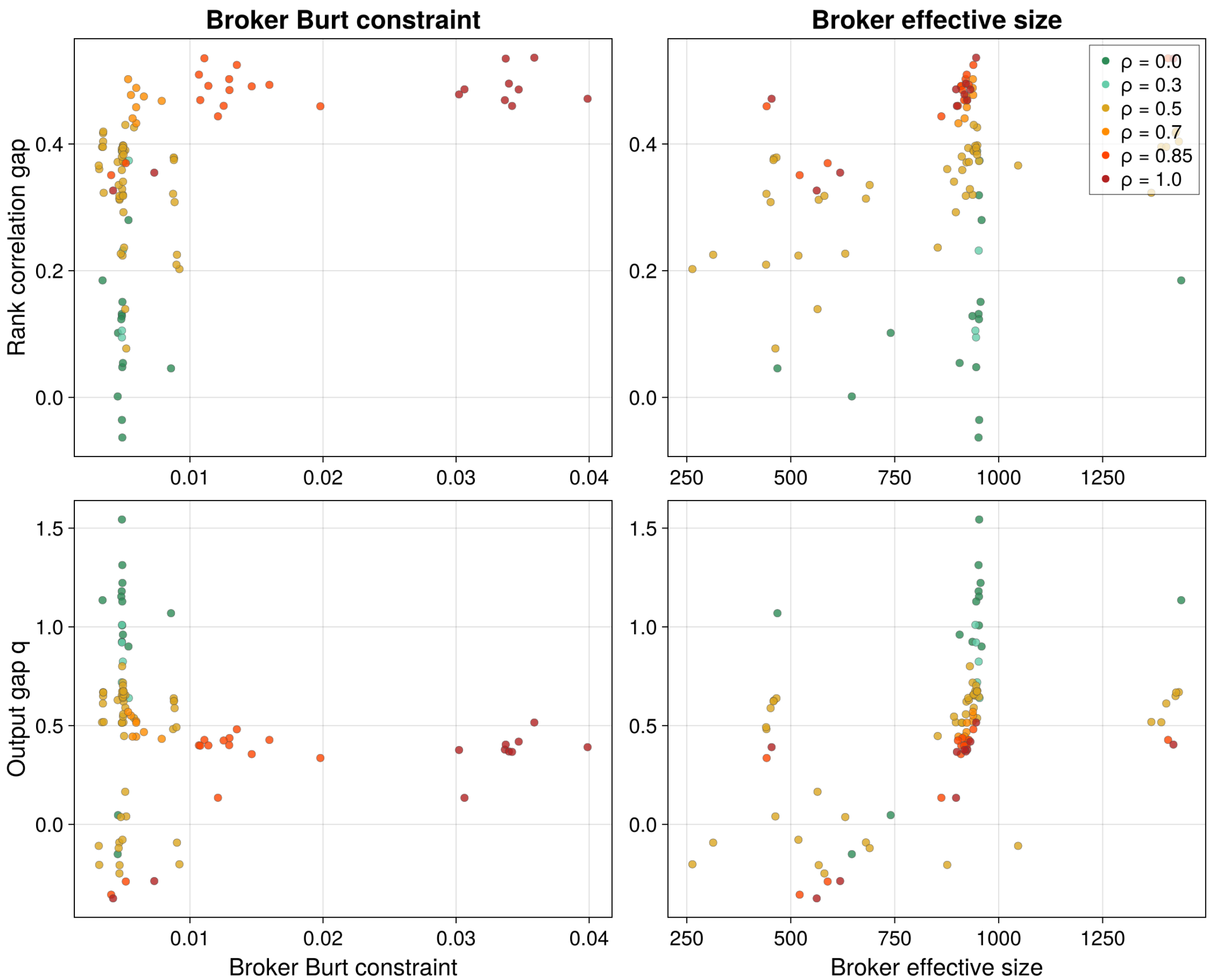

Figure S3: Rank-correlation gap (broker minus principals, top) and output gap (broker channel minus self-search, bottom) against broker Burt constraint (left) and broker effective size (right); one dot per regime, each a late-window mean over that regime's seeds, colored by the matching parameter ρ.

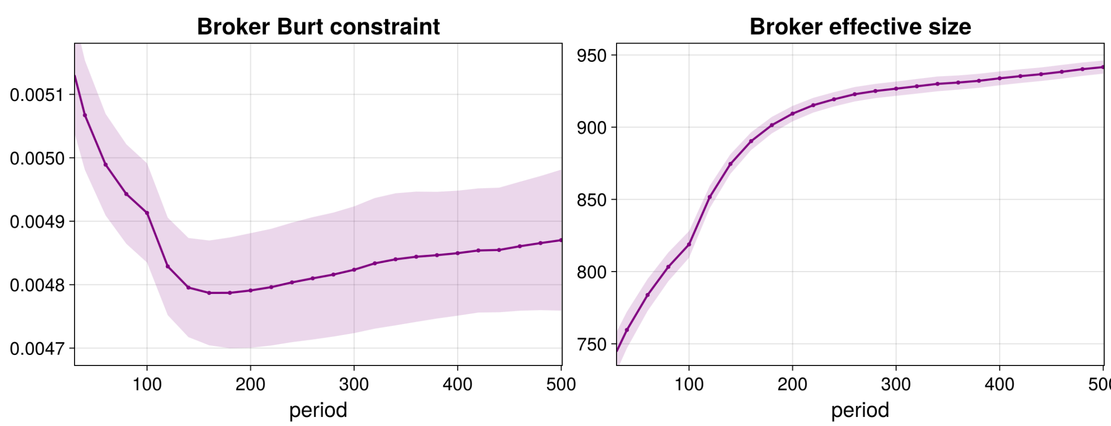

Figure S4: Broker Burt constraint (left) and broker effective size (right) over time at the baseline regime, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals, measured every 20 periods. Axes start at $t = 30$ to omit the initialization transient.

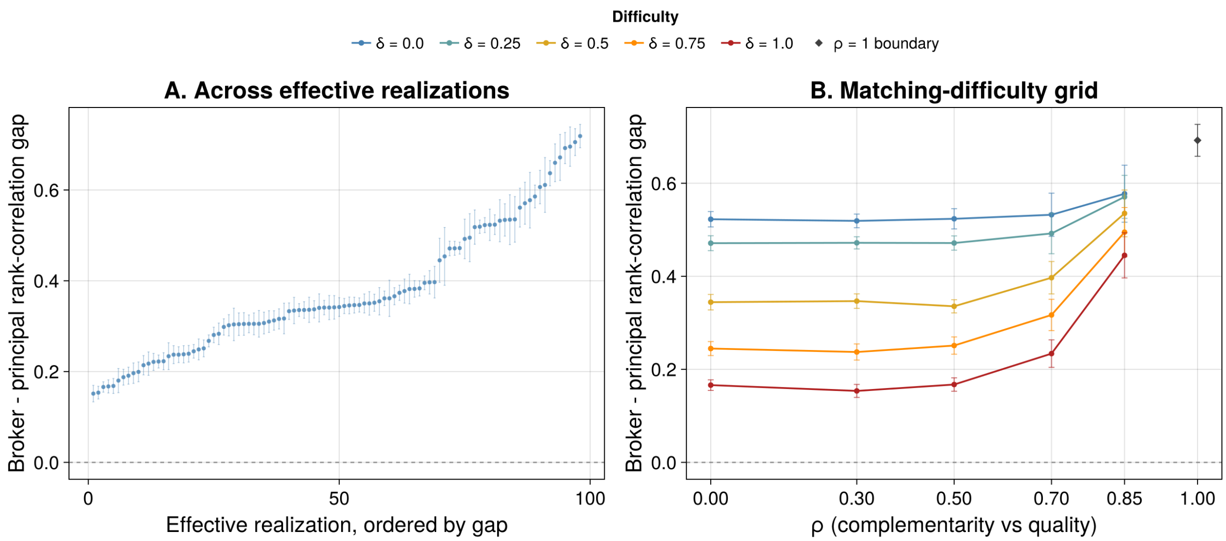

Figure S5: Broker ranking advantage under paired-Ridge learning. Panel A orders the 98 regimes by their late-period broker-minus-principal rank-correlation gap. The gap is positive in 98 regimes and has a median of 0.342; points are regime means and whiskers are 95% Monte Carlo intervals over seeds. Panel B shows the same Ridge gap across the $\rho \times \delta$ design, with one line per value of δ over $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are regime means and whiskers are 95% Monte Carlo intervals. The baseline gap is 0.335.